

Supplementary Information for:
COACH: Toward the performance limit of
traditional hybrid density functionals for molecular
chemistry

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This PDF file includes:

- Full theoretical derivations of the functional form
- Detailed description of constraints and their enforcement
- Expanded methods for optimization and numerical procedures
- Detailed design exploration and tests
- Further comments on nonlocal functional

1 General Functional Form

In this work, we adopt a generalized version of the B97-type functional form [1, 2] within the range-separated hybrid (RSH) meta-generalized-gradient-approximation (mGGA) framework, similar to that employed in the ω B97M-V functional [3]. The total exchange energy (E_x) comprises three components: short-range (sr) Hartree-Fock (HF) exchange, long-range (lr) HF exchange, and semi-local (SL) exchange. The total correlation energy (E_c) is also partitioned into SL same-spin (ss) correlation, SL opposite-spin (os) correlation, and a dispersion correction term (disp).

$$E_x = E_{x, sr}^{SL} + c_{x, sr}^{HF} E_{x, sr}^{HF} + c_{x, lr}^{HF} E_{x, lr}^{HF}, \quad (1)$$

$$E_c = E_{c,ss}^{SL} + E_{c,os}^{SL} + E_c^{disp}. \quad (2)$$

In this framework, $E_{x,sr}^{HF}$ and $E_{x,lr}^{HF}$ are defined as

$$E_{x,sr}^{HF} = -\frac{1}{2} \sum_{\sigma=\alpha,\beta} \sum_{i,j}^{n_\sigma} \int \int \phi_{i\sigma}^*(\mathbf{r}_1) \phi_{j\sigma}^*(\mathbf{r}_2) \frac{\text{erfc}(\omega|\mathbf{r}_1 - \mathbf{r}_2|)}{|\mathbf{r}_1 - \mathbf{r}_2|} \phi_{j\sigma}(\mathbf{r}_1) \phi_{i\sigma}(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2, \quad (3)$$

$$E_{x,lr}^{HF} = -\frac{1}{2} \sum_{\sigma=\alpha,\beta} \sum_{i,j}^{n_\sigma} \int \int \phi_{i\sigma}^*(\mathbf{r}_1) \phi_{j\sigma}^*(\mathbf{r}_2) \frac{\text{erf}(\omega|\mathbf{r}_1 - \mathbf{r}_2|)}{|\mathbf{r}_1 - \mathbf{r}_2|} \phi_{j\sigma}(\mathbf{r}_1) \phi_{i\sigma}(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2, \quad (4)$$

The Coulomb operator is split into long-range and short-range components using the error function and complementary error function, respectively. Here ω is the range-separation parameter controlling the transition from short-range to long-range exchange.

In our functional development, we set $c_{x,lr}^{HF} = 1$ as the default, meaning the nonlocal HF exchange is fully recovered for long-range interactions. Such functionals are called ‘long-range-corrected’ (LRC) functionals.[4] The dispersion correction E_c^{disp} may be implemented as an empirical DFT-D correction like D4 [5] or a nonlocal correlation functional, such as VV10 [6].

The semi-local exchange and correlation terms ($E_{x,sr}^{SL}$, $E_{c,ss}^{SL}$, $E_{c,os}^{SL}$) share a similar general functional form,

$$E_{x(c)}^{SL} = \int e_{x(c)}^{\text{base}} F_{\text{corr}} g_{x(c)} d\mathbf{r}, \quad (5)$$

where:

- $e_{x(c)}^{\text{base}}$ is the exchange (or correlation) base energy density, which is usually chosen as a form that exactly models the uniform electron gas (UEG).
- F_{corr} is a multiplicative correction factor with physical significance, often used to satisfy certain physical constraints.
- $g_{x(c)}$ is an additional correction factor that must be determined by optimization against reference data.

Together, the product of F_{corr} and $g_{x(c)}$ forms the inhomogeneity correction factors (ICFs), which capture the effects of density inhomogeneity on the exchange and correlation energies. The SL exchange and correlation components of different functionals are usually designed with similar base functionals and always with different ICFs. Let us elaborate on this in more detail.

1.1 Base Energy Density

1. **Short-Range Exchange Energy** E_x^{sr} : The exact form of the UEG exchange energy density per unit volume is known,

$$e_{x,\sigma}^{\text{UEG}} = -\frac{3}{2} \left(\frac{3}{4\pi} \right)^{1/3} \rho_\sigma^{4/3}, \quad (6)$$

which is typically used as the exchange base energy density. To transform the exchange functional to its short-range counterpart suitable for RSH,[7] a multiplicative factor $F_{x,\sigma}^{\text{sr}}$ is introduced [4, 8],

$$F_{x,\sigma}^{\text{sr}} = 1 - \frac{2}{3}a_\sigma \left(2\sqrt{\pi} \operatorname{erf} \left(\frac{1}{a_\sigma} \right) - 3a_\sigma + a_\sigma^3 + (2a_\sigma - a_\sigma^3) \exp \left(-\frac{1}{a_\sigma^2} \right) \right), \quad (7)$$

where $a_\sigma = \frac{\omega}{k_{F_\sigma}}$, $k_{F_\sigma} = (6\pi^2\rho_\sigma)^{1/3}$ is the spin-polarized Fermi wave vector, and ω is the range-separation parameter in Eqn 4.

2. **Same-Spin Correlation Energy E_c^{ss}** : The same-spin correlation base energy density is derived from the base correlation,

$$e_{c,\sigma\sigma}^{\text{base}} = e_c^{\text{base}}(\rho_\sigma, 0). \quad (8)$$

Because the exact form of the UEG correlation energy is not known, a specific fit in the form of the local spin density approximation, such as PW92 [9], is commonly used as the base. However, PW92 does not satisfy several physical constraints, as discussed in the next section. Therefore, we also consider the SCAN correlation functional (at $\alpha = 1$, see actual implementation in SI of Ref. 10) as the base.

3. **Opposite-Spin Correlation Energy E_c^{os}** : The opposite-spin correlation base energy density is similarly derived from the base correlation,

$$e_{c,os}^{\text{base}} = e_c^{\text{base}}(\rho_\alpha, \rho_\beta) - e_c^{\text{base}}(\rho_\alpha, 0) - e_c^{\text{base}}(0, \rho_\beta). \quad (9)$$

1.2 Dimensionless Variables

For a meta-GGA, the fitting factors $g_{x(c)}$ depend on the density gradient $\nabla\rho_\sigma$ and the kinetic energy density τ_σ . To facilitate fitting and satisfy physical constraints, their appropriate corresponding dimensionless variables must be introduced.

The most common dimensionless variable that represents the density gradient $\nabla\rho_\sigma$ is s_σ , which is transformed into $u_{x,\sigma}$, $u_{c,\sigma\sigma}$, and $u_{c,\alpha\beta}$ via finite-domain transformations to simplify fitting and avoid divergence.

$$s_\sigma = \frac{|\nabla\rho_\sigma|}{\rho_\sigma^{4/3}}, \quad u_{x,\sigma} = \frac{\gamma_x s_\sigma^2}{1 + \gamma_x s_\sigma^2}, \quad u_{c,\sigma\sigma} = \frac{\gamma_{c,ss} s_\sigma^2}{1 + \gamma_{c,ss} s_\sigma^2}, \quad (10)$$

$$s_{\alpha\beta}^2 = \frac{1}{2}(s_\alpha^2 + s_\beta^2), \quad u_{c,\alpha\beta} = \frac{\gamma_{c,os} s_{\alpha\beta}^2}{1 + \gamma_{c,os} s_{\alpha\beta}^2}, \quad (11)$$

Here, γ_x , $\gamma_{c,ss}$, and $\gamma_{c,os}$ are nonlinear parameters determined through fitting. In this work, we adopt $\gamma_x = 0.004$, which was fitted to the Hartree-Fock exchange energies of 20 atoms in 1986 by Becke [11]. For $\gamma_{c,os}$, we use 0.006, which was fitted to the correlation energies of neon and helium in 1997 by Becke [1]. For $\gamma_{c,ss}$, Matito et al. recently demonstrated that the value of 0.2 used in the B97 functional is too large and causes oscillations in the functional derivative along potential energy curves[12].

Therefore, we tested 0.005, 0.01, 0.02, and 0.04, and selected 0.01 because it is the smallest value that does not significantly impair the accuracy of the resulting functional.

A number of dimensionless variables involving the kinetic energy density $\tau_\sigma = \sum_{i=1}^{n_\sigma} |\nabla \phi_{i,\sigma}|^2$ have been proposed, including

$$t_\sigma = \frac{\tau_\sigma^{\text{UEG}}}{\tau_\sigma}, \quad (12)$$

$$z_\sigma = \frac{\tau_\sigma^W}{\tau_\sigma}, \quad (13)$$

$$\alpha_\sigma = \frac{\tau_\sigma - \tau_\sigma^W}{\tau_\sigma^{\text{UEG}}}, \quad (14)$$

$$\beta_\sigma = \frac{\tau_\sigma - \tau_\sigma^W}{\tau_\sigma + \tau_\sigma^{\text{UEG}}}. \quad (15)$$

Here, the UEG kinetic energy density τ_σ^{UEG} and the von Weizsäcker kinetic energy density τ_σ^W are defined as

$$\tau_\sigma^{\text{UEG}} = \frac{3}{5} (6\pi^2)^{2/3} \rho_\sigma^{5/3}, \quad (16)$$

$$\tau_\sigma^W = \frac{1}{4} \frac{|\nabla \rho_\sigma|^2}{\rho_\sigma}, \quad (17)$$

which represent the UEG limit and the single-orbital limit of τ_σ , respectively. These prefactors follow the Q-Chem convention for τ_σ given above, which is twice the convention that includes a factor of 1/2. As shown in Table 1, these dimensionless variables take different values in different chemical environments. Among them, t_σ and α_σ are the most frequently employed in functional development. However, t_σ cannot distinguish single-orbital regions and α_σ has known numerical instabilities. The β_σ variable aims to mitigate the numerical problems of α_σ while retaining its ability to distinguish single-orbital regions [13]. To our knowledge, no functional has employed β_σ to date. We have explored replacing t_σ with the more expressive variable β_σ , and decided to adopt β_σ for exchange and same-spin correlation terms where it appears to facilitate improved accuracy.

Table 1 Common dimensionless variables involving the kinetic energy density.

Region	t	z	α	β
Single orbital	$\frac{3}{5s^2}$	1	0	0
Slowly varying density	≈ 1	≈ 0	≈ 1	$\approx \frac{1}{2}$
Noncovalent bonding	≈ 0	≈ 0	$\gg 1$	$\frac{1}{2} \ll \beta < 1$
Representative functional	B97M-V [14]	TPSS [15]	SCAN [10]	This work

1.3 Fitting Factor

We choose to expand $g_{x\sigma}$ and $g_{c,\sigma\sigma}$ as a series in terms of the two chosen independent variables, $u_{x,\sigma}$ and β_σ , after transforming β_σ from $[0,1]$ to $[-1,1]$. Taking $g_{x\sigma}$ as an example, we have:

$$g_{x\sigma} = g_{x\sigma}(\nabla\rho_\sigma, \tau_\sigma) = \sum_{i=0}^M \sum_{j=0}^N c_{x,ij} f_i(2\beta_\sigma - 1) f_j(u_{x,\sigma}), \quad (18)$$

where $f_i(x)$ are the expansion functions (to be specified below). M and N are the maximum expansion orders of $2\beta_\sigma - 1$ and $u_{x,\sigma}$, respectively.

Similarly, we define $g_{c,os}$ to be a function of two independent variables $u_{c,\alpha\beta}$ and $t_{\alpha\beta}$, and expand it as

$$g_{c,os} = g_{x\sigma}(\nabla\rho_\sigma, \tau_\sigma) = \sum_{i=0}^M \sum_{j=0}^N c_{x,ij} f_i(w_{c,\alpha\beta}) f_j(u_{c,\alpha\beta}) \quad (19)$$

$$w_{c,\alpha\beta} = \frac{t_{\alpha\beta} - 1}{t_{\alpha\beta} + 1}, \quad t_{\alpha\beta} = \frac{1}{2}(t_\alpha + t_\beta)$$

where again $f_i(x)$ are the expansion functions, and M and N are the maximum expansion orders of $w_{c,\alpha\beta}$ and $u_{x,\sigma}$, respectively.

Most previous B97-type functionals have used monomial basis (i.e., $f_i(x) = x^i$). The CASE functional uniquely employs B-splines and bell-shaped spline functions, yielding encouraging results [16]. In this work, we investigate the use of Legendre polynomials and Chebyshev polynomials as expansion functions due to their smoothness and the ease with which they can be converted into monomial basis.

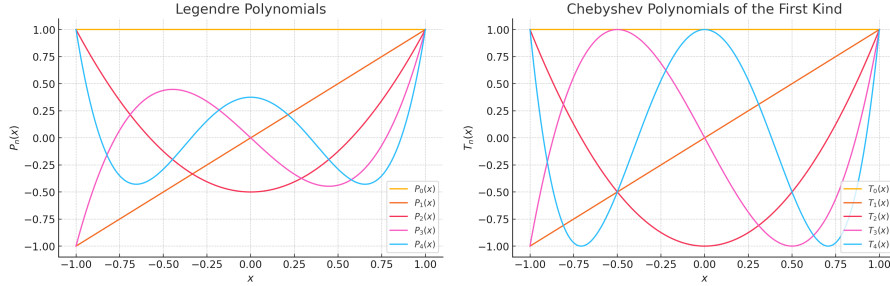


Fig. 1 Plot of the Legendre polynomials $P_n(x)$ and Chebyshev polynomials $T_n(x)$ for $n = 0$ to 4

Legendre polynomials $P_n(x)$ are a sequence of orthogonal polynomials defined on the interval $[-1, 1]$, which are solutions to Legendre's differential equation. They are defined recursively by

$$P_0(x) = 1, \quad P_1(x) = x,$$

$$P_{n+1}(x) = \frac{2n+1}{n+1}xP_n(x) - \frac{n}{n+1}P_{n-1}(x), \quad n \geq 1. \quad (20)$$

The Legendre polynomials are orthogonal,

$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1}\delta_{mn}, \quad (21)$$

where δ_{mn} is the Kronecker delta function, equal to 1 when $m = n$ and 0 otherwise. Legendre polynomials can form a complete basis set for target functions on $[-1, 1]$. This makes them useful in expanding functions in series that converge rapidly, which is advantageous in numerical computations and fitting procedures.

Chebyshev polynomials of the first kind $T_n(x)$ are another sequence of orthogonal polynomials defined on the interval $[-1, 1]$, with orthogonality weight $w(x) = (1 - x^2)^{-1/2}$. They are defined recursively by

$$T_0(x) = 1, \quad (22)$$

$$T_1(x) = x, \quad (23)$$

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \quad n \geq 1. \quad (24)$$

Important properties of Chebyshev polynomials include their orthogonality,

$$\int_{-1}^1 \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}} dx = \begin{cases} 0, & m \neq n, \\ \frac{\pi}{2}, & m = n \neq 0, \\ \pi, & m = n = 0. \end{cases} \quad (25)$$

They also satisfy the identity,

$$T_n(\cos \theta) = \cos(n\theta), \quad (26)$$

which links them directly to trigonometric functions and facilitates efficient computational algorithms. Chebyshev polynomials are highly effective in polynomial approximations and spectral methods, which might be helpful for developing smooth and accurate functional expansions in density functional theory.

2 Constraints

The SCAN functional was designed to satisfy all 17 known exact constraints that a meta-GGA functional could fulfill at the time of its development [10]. In developing our new RSH mGGA functional, we aim to first assess these constraints and determine whether they can be satisfied by ω B97M-V and our new COACH functional. Additionally, we introduce the numerical stability constraint, which is important for practical application of functionals. We also mention the constraints unsatisfied by semi-local functionals, which give rise to the main errors associated with practical density functionals.

2.1 Exchange Constraints

(1) *Negativity:*

The exchange energy must always be negative.

Assessment: Because the UEG and both HF exchange terms are non-positive and the HF coefficients are nonnegative, $G_{x,\sigma} = F_{x,\sigma}^{\text{NUSC}} g_{x,\sigma} \geq 0$ is sufficient.

(2) *Spin-Scaling:*

The exchange energy satisfies the spin-scaling relation,

$$E_x(\rho_\alpha, \rho_\beta) = \frac{1}{2}E_x(\rho_\alpha, \rho_\alpha) + \frac{1}{2}E_x(\rho_\beta, \rho_\beta). \quad (27)$$

Assessment: This condition is straightforward to satisfy since both ω B97M-V and COACH employ the additive spin-density form $E_x = \sum_{\sigma}^{\alpha,\beta} E_{x,\sigma}(\rho_\sigma)$, where $E_{x,\sigma}$ only depends on the density variables with spin σ .

(3) *Uniform Density Scaling:*

Under uniform density scaling [17], the exchange energy should scale linearly,

$$E_x(\rho_\lambda) = \lambda E_x(\rho), \quad (28)$$

where $\rho_\lambda(\mathbf{r}) = \lambda^3 \rho(\lambda \mathbf{r})$.

Assessment: For the LSDA,

$$\begin{aligned} E_{x,\sigma}(\rho_\lambda) &= \int -\frac{3}{2} \left(\frac{3}{4\pi} \right)^{1/3} (\rho_{\lambda,\sigma}(\mathbf{r}))^{4/3} d\mathbf{r} \\ &= \int -\frac{3}{2} \left(\frac{3}{4\pi} \right)^{1/3} \lambda^4 \rho_\sigma(\lambda \mathbf{r})^{4/3} d\mathbf{r} \\ &= \lambda \int -\frac{3}{2} \left(\frac{3}{4\pi} \right)^{1/3} \rho_\sigma(\lambda \mathbf{r})^{4/3} d(\lambda \mathbf{r}) \\ &= \lambda E_{x,\sigma}(\rho), \end{aligned} \quad (29)$$

satisfies this constraint. The dimensionless variables s_σ , t and β_σ remain unchanged under uniform density scaling. Therefore, this constraint can be satisfied by any global hybrid B97-type functional.

However, the short-range exchange correction factor $F_{x,\sigma}^{\text{sr}}$ does not fully satisfy this constraint due to the fixed ω . When we scale the density uniformly, such that $\rho \rightarrow \rho_\lambda$, the electron density becomes either concentrated ($\lambda > 1$) or diluted ($\lambda < 1$). However, a fixed ω cannot match the effective range-separation scale of the modified density and thus the range-separation behavior is disrupted. Strictly speaking, RSH functionals cannot exactly satisfy this constraint. However, there is a lucky compensating effect: the range-separation of the exact HF exchange energy will be disrupted in a similar way. Consequently, the total exchange energy, which includes both the (lr and sr) HF

and semi-local exchanges, will approximately satisfy the scaling constraint, and the deviation from the exact scaling behavior will be roughly proportional to the difference between the exact short-range HF exchange and approximated semi-local exchange.

(4) Fourth-Order Gradient Expansion (GE4):

In the limit of slowly varying densities ($s \ll 1$ and $\alpha \approx 1$), the ICF should reproduce the fourth-order gradient expansion [18],

$$g_x^{\text{GE4}}(s, \alpha) = 1 + \frac{10}{81}s^2 - \frac{1606}{18225}s^4 + \frac{511}{13500}s^2(1 - \alpha) + \frac{5913}{405000}(1 - \alpha)^2, \quad (30)$$

where s is defined in the SCAN functional as $s = \frac{|\nabla\rho|}{2(3\pi^2)^{1/3}\rho^{4/3}}$, which differs by a constant and spin summation from the B97-type s definition. There are also two less restrictive constraints:

(4a) Zeroth-Order Approximation: The functional should exactly reproduce the UEG exchange in the uniform limit, where $g_x = 1$ when $\nabla\rho = 0$ everywhere.

(4b) Second-order Gradient Expansion (GE2) [19]: In the limit of slowly varying densities ($s \ll 1$), $g_x^{\text{GE2}}(s) = 1 + \frac{10}{81}s^2$.

Assessment: $\omega\text{B97M-V}$ and COACH can satisfy constraint (4a) by setting $c_{x,\text{lr}}^{\text{HF}} = 1$ and $g_{x,\sigma}^{\text{UEG}} + c_{x,\text{sr}}^{\text{HF}} = 1$; for COACH , $g_{x,\sigma}^{\text{UEG}} = 0.771210192833593 = 1 - c_{x,\text{sr}}^{\text{HF}}$. However, we will not enforce the more restrictive (GE2 and GE4) constraints for two reasons. First, ensuring exact compliance with GE4 could significantly restrict functional flexibility, likely reducing performance. Second, if we only enforce the GE2, the second-order parameter is shown to be unsuitable for finite systems like molecules, with empirical coefficients often two to three times the theoretical estimates from linear response theory [11, 20, 21]. For example, $\omega\text{B97M-V}$, which performs exceptionally well in molecular energy predictions, does not satisfy this constraint.

(5) Non-Uniform Density Scaling:

Under non-uniform density scaling, the exact exchange energy per electron remains finite [22, 23],

(5a) towards the high-density limit (two-dimensional limit),

$$\lim_{\lambda \rightarrow \infty} \frac{1}{N} E_x(\rho_\lambda^x) > -\infty, \quad (31)$$

(5b) towards the low density limit,

$$\lim_{\lambda \rightarrow 0} \frac{1}{N} E_x(\rho_\lambda^x) > -\infty, \quad (32)$$

where $\rho_\lambda^x(x, y, z) = \lambda\rho(\lambda x, y, z)$.

Assessment: Since our dimensionless variables are bounded within a finite domain, the fitting factor $g_{x,\sigma}$ will remain finite under the scaling. Then the UEG exchange energy (base) scale as $\lambda^{1/3}$, which reaches zero as $\lambda \rightarrow 0$ but diverges when $\lambda \rightarrow \infty$. Therefore, $\omega\text{B97M-V}$ and COACH can both satisfy (5b) but $\omega\text{B97M-V}$ cannot satisfy

(5a). The SCAN functional satisfies (5a) by introducing a non-uniform scaling correction (NUSC) factor that behaves as $F_x \sim s^{-1/2}$ when $s \rightarrow \infty$ to cancel the $\lambda^{1/3}$ behavior of UEG exchange energy density. We try to incorporate SCAN's NUSC factor in COACH to ensure compliance with the constraint (5a). The NUSC factor in our work is defined as

$$g_x(s_\sigma) = 1 - \exp[-a_1 s_\sigma^{-1/2}] \quad (33)$$

Here $a_1 = 13.815$, adopted from SCAN's a_1 value (adjusted due to different s^2 definitions). Since $0 \leq \text{erf}, \text{erfc} \leq 1$, each HF range term is bounded by full exact exchange, so finite coefficients in hybrid functionals preserve these limits.

(6) Tight Bound for Spin-Unpolarized Two-Electron Densities:

For spin-unpolarized two-electron densities, the exchange enhancement factor should satisfy [24–26]

$$F_x \leq 1.174 \quad \text{for } E_x = \int e_x^{UEG} F_x d\mathbf{r}. \quad (34)$$

Assessment: This is a constraint that is not met by ω B97M-V, but the semi-local exchange component of our functional is designed to satisfy it numerically. For a spin-unpolarized two-electron density, we have $\beta_\sigma = 0$ (or $t_\sigma^{-1} = \frac{5s_\sigma^2}{12(6\pi^2)^{2/3}}$ when using t). We can enforce $g_{x,\sigma} \leq 1.174 \times 2^{1/3} = 1.479$ on the mesh grids of the line $\beta_\sigma = 0$ (or $t_\sigma^{-1} = \frac{5s_\sigma^2}{12(6\pi^2)^{2/3}}$) in the optimization. However, the final RSH exchange can not strictly satisfy this constraint because neither $E_{x,\text{sr}}^{\text{HF}} \geq 1.174 E_{x,\text{sr}}^{\text{LDA}}$ nor $E_{x,\text{lr}}^{\text{HF}} \geq 1.174 E_{x,\text{lr}}^{\text{LDA}}$ is guaranteed.

2.2 Correlation Constraints

(7) Non-Positivity:

The correlation energy is non-positive [17].

Assessment: This constraint can be met by enforcing $g_{c,ss} \geq 0$ and $g_{c,os} \geq 0$ during fitting. However, ω B97M-V does not satisfy this constraint, benefiting instead from the potential exchange-correlation error cancellation. We will assess the impact of this constraint and determine whether it should be applied in COACH.

(8) Second-Order Gradient Expansion for Correlation:

In the slowly varying limit ($t \rightarrow 0$), the correlation energy should reproduce the second-order gradient expansion [27] approximated as

$$H \rightarrow \beta \phi^3 t^2, \quad (35)$$

Here, H is the additional energy added to LDA correlation energy density. See the definition of other variables in Ref 28.

Assessment: This constraint is difficult to implement, as it requires an additive correction to the LDA correlation energy, whereas ω B97M-V and our functional use a multiplicative correction. Considering flexibility requirements, we will not enforce this constraint in our COACH functional.

(9) Uniform Density Scaling to the High-Density Limit:

Under uniform density scaling to the high-density limit ($\lambda \rightarrow \infty$) [17, 22],

$$\lim_{\lambda \rightarrow \infty} E_c(\rho_\lambda) > -\infty, \quad (36)$$

and

$$E_c(\rho_\lambda) > \lambda E_c(\rho), \quad \lambda > 1. \quad (37)$$

Assessment: This constraint requires cancellation of the logarithmic singularity present in the LDA correlation energy (e.g., PW92 [9]). Implementing this correction in our functional is non-trivial due to the multiplicative form of our correlation energy expression. We can satisfy this constraint by changing our base correlation functional from PW92 to SCAN ($\alpha = 1$).

(10) Uniform Density Scaling to the Low-Density Limit:

In the low-density limit ($\lambda \rightarrow 0$) [17, 29],

$$\lim_{\lambda \rightarrow 0} E_c(\rho_\lambda) = \lambda D_c(\rho), \quad (38)$$

where $D_c(\rho)$ is an unknown density functional, and

$$E_c(\rho_\lambda) < \lambda E_c(\rho), \quad \lambda < 1. \quad (39)$$

Assessment: This constraint is satisfied by both PW92 and SCAN correlation functional. Since t and s (or α) remain unchanged under uniform density scaling, both ω B97M-V and COACH meet this constraint.

(11) Zero Correlation Energy for One-Electron Density:

The correlation energy should be zero for a one-electron density.

Assessment: For fully spin-polarized systems, our opposite-spin correlation base energy density is zero by definition. Therefore, we only need to add a self-correlation correction (SCC) factor $F_{c,ss}^{\text{SCC}}$, which is exactly zero for one-electron systems, to make same-spin correlation energy zero. ω B97M-V does not have this SCC factor, and we will assess the effect on our COACH functional.

The most popular self-correlation correction factor $\alpha_\sigma = \frac{\tau_\sigma - \tau_\sigma^{\text{W}}}{\tau_\sigma}$ is shown to be numerically unstable [30]. Therefore, we propose a new numerically stable SCC factor $2\beta_\sigma$ that equals 0 for one-electron systems and is equal to 1 for the UEG.

(12) Non-Uniform Density Scaling:

Under non-uniform density scaling [23, 31], with $\rho_\lambda^x(x, y, z) = \lambda\rho(\lambda x, y, z)$,

(12a) towards the high density limit,

$$\lim_{\lambda \rightarrow \infty} \frac{1}{N} E_c(\rho_\lambda^x) > -\infty. \quad (40)$$

(12b) towards the low-density limit,

$$\lim_{\lambda \rightarrow 0} E_c(\rho_\lambda^x) = 0. \quad (41)$$

Assessment: The PW92 functional can satisfy (12b) as it tends to zero as $\lambda \rightarrow 0$. However, it diverges logarithmically as $\lambda \rightarrow \infty$. The multiplicative form does not allow the necessary cancellation of divergences in the high-density limit. This problem can be solved by changing our base correlation functional from PW92 to SCAN ($\alpha = 1$).

2.3 Exchange and Correlation Constraints

(13) *Size Extensivity:*

As for the total energy, the exchange-correlation energy should be size-extensive [32].

Assessment: The semi-local and both HF range terms are additive for well-separated noninteracting systems, so fixed mixing coefficients preserve size extensivity.

(14) *Tighter General Lieb-Oxford Bound (Chan-Handy Bound):*

The exchange-correlation energy must satisfy the tighter Lieb-Oxford bound [33, 34]:

$$E_{xc}(\rho) \geq -L_{CH} \int \rho^{4/3}(\mathbf{r}) d\mathbf{r}, \quad (42)$$

where $L_{CH} = 2.2146$.

Assessment: Both ω B97M-V and COACH satisfy the corresponding bound for their semi-local exchange components, but neither complete RSH functional is guaranteed to satisfy XC14 because the analogous SR and LR HF-LDA bounds are not established.

(15) *Weak Dependence on Spin Polarization in the Low-Density Limit:*

In the low-density limit, the leading exact exchange-correlation energy is independent of the relative spin polarization ξ [9, 10].

Assessment: In the unmodified LSDA, the spin dependence of the PW92 low-density correlation coefficient largely cancels that of LSDA exchange, leaving only a weak residual spin dependence in the total exchange-correlation coefficient [9]. Standard SCAN is constructed to satisfy the corresponding low-density spin-independence constraint [10]. However, this is a constraint on the total exchange-correlation energy and is not inherited merely by using PW92 or SCAN as a correlation base. Both ω B97M-V and COACH decompose the base correlation into same-spin and opposite-spin contributions and multiply these contributions by distinct B97-type inhomogeneity correction factors; their range-separated mixtures of semilocal and Hartree-Fock exchange further alter the exchange-correlation balance. Consequently, the cancellation required by XC15 is not enforced in either functional, and neither ω B97M-V nor COACH strictly satisfies this constraint.

(16) Static Linear Response of the Uniform Electron Gas:

The functional should reproduce the correct linear response behavior of the uniform electron gas.

Assessment: This constraint is the consequence of obeying the constraints (4) and (8). As ω B97M-V does not satisfy the static linear response of the UEG, we will not enforce it in COACH, either.

(17) Lieb-Oxford Bound for Two-Electron Densities:

For two-electron systems, the exchange-correlation energy must satisfy a Lieb-Oxford bound with a constant greater than or equal to 1.671 [24].

Assessment: Since we lack an indicator specific to two-electron spin-polarized systems, we cannot satisfy this constraint easily. ω B97M-V does not obey it either.

2.4 Summary

In summary, ω B97M-V satisfies constraints (1), (2), (10), and (13), totaling four constraints, and partially satisfies five others: (3), (4a), (5b), (12b), and (14). As demonstrated below and in the main text, COACH is constructed to fulfill constraints (2), (5), (9), (10), (11), (12), and (13), totaling seven constraints, and to partially, on the sampled mesh, or at the component level satisfy constraints (1), (3), (4a), (6), and (14). By contrast, constraint (4), the Fourth-Order Gradient Expansion (GE4) for exchange, is not fully enforced, while constraints (7), non-positivity of correlation; (8), the second-order gradient expansion for correlation; (15), low-density spin independence; (16), the static linear response of the uniform electron gas; and (17), the Lieb-Oxford bound for two-electron densities, are not enforced. Within the framework of the impossible triangle, these constraints are infeasible to impose without compromising performance, given our adoption of the RSH meta-GGA functional form.

Finally it is worth mentioning that the two main errors in approximate density functionals themselves arise from other constraints that are not satisfied (again due to limitations of the RSH meta-GGA functional form). The delocalization or self-interaction error (SIE) arises from the failure of approximate exchange functionals to cancel the classical Coulomb repulsion energy in 1-electron systems. The strong correlation error arises from the additional failure of the XC functional to be exact for clusters of two or more electrons.

2.5 Additional Grid Sensitivity Constraint

Minimizing grid-related numerical errors is crucial for practical functional use. SCAN’s failure to be numerically stable demonstrates the need for grid sensitivity constraints. ω B97M-V controls the energy difference between grids (99,590)/SG-1 ((99,590) for semi-local xc part and SG-1 for VV10 part) and (250,974)/SG-1 to within 0.015 kcal/mol [35], allowing it to maintain adequate grid robustness even for second-order properties [36]. We will continue to enforce this constraint and monitor grid-based errors during functional development.

3 Optimization Protocol

3.1 Weighted Least-Squares Fitting Problem

To determine the optimal expansion coefficients $c_{x(c),ij}$ in our functional form, we formulate the parameter optimization as a linear least-squares fitting problem. This approach allows us to fit the exchange and correlation energies calculated from our functional to a set of reference data, minimizing the overall deviation.

For a given set of training data comprising K systems or properties (e.g., total energies, reaction energies, dipole moments, etc.), we aim to minimize the loss function L , defined as the sum of weighted squared deviations between the calculated and reference values,

$$L = \sum_{k=1}^K \frac{1}{2} w_k [E_k^{\text{calc}} - E_k^{\text{ref}}]^2, \quad (43)$$

where E_k^{calc} is the reaction energy calculated using COACH for data point k , E_k^{ref} is the corresponding reference energy from high-level calculations or experimental data, w_k is a weight assigned to data point k , reflecting its importance in the fitting procedure.

To simplify the optimization into a linear least-squares problem, we construct the COACH energy based on the energy (E^{base}) and fixed density of a base functional,

$$E^{\text{calc}} = E^{\text{base}} + \Delta E_{xc}. \quad (44)$$

For example, using $\omega\text{B97X-V}$ with range-separation parameter ω_0 as the base functional and retaining the same VV10 energy as the dispersion correction, we have

$$\begin{aligned} \Delta E_{xc} &= E_{xc}^{\text{calc}} - E_{xc}^{\omega\text{B97X-V}} \\ &= E_{x,\text{sr}}^{\text{SL,calc}}(\omega) + c_{x,\text{sr}}^{\text{HF,calc}} E_{x,\text{sr}}^{\text{HF}}(\omega) + E_{x,\text{lr}}^{\text{HF}}(\omega) + E_{c,\text{ss}}^{\text{SL,calc}} + E_{c,\text{os}}^{\text{SL,calc}} \\ &\quad - (E_{x,\text{sr}}^{\text{SL},\omega\text{B97X-V}}(\omega_0) + c_{x,\text{sr}}^{\text{HF},\omega\text{B97X-V}} E_{x,\text{sr}}^{\text{HF}}(\omega_0) + E_{x,\text{lr}}^{\text{HF}}(\omega_0) \\ &\quad + E_{c,\text{ss}}^{\text{SL},\omega\text{B97X-V}} + E_{c,\text{os}}^{\text{SL},\omega\text{B97X-V}}). \end{aligned} \quad (45)$$

The coefficient of long-range HF exchange is fixed at $c_{x,\text{lr}}^{\text{HF}} = 1$ for both functionals. Thus, the long-range term is not an adjustable feature in the coefficient vector. When $\omega \neq \omega_0$, its explicitly recomputed change enters the loss through the target energy defined below. Here, the SL terms can be expressed generally as

$$E^{\text{SL,calc}} = \sum_{i=0}^M \sum_{j=0}^N c_{x(c),ij} F_{x(c),ij} \quad (46)$$

$$F_{x(c),ij} = \int e_{x(c)}^{\text{base}} F_{\text{corr}} f_i(2\beta_\sigma - 1) f_j(u_{x(c),\sigma}) d\mathbf{r}. \quad (47)$$

This allows us to rewrite the loss function as

$$L = \sum_{k=1}^K \frac{1}{2} w_k \left[\sum_{l=1}^L c_l F_{k,l} - E_k^{\text{To-fit}} \right]^2, \quad (48)$$

where:

- c_l is the l -th linear coefficient, obtained by flattening the two-dimensional indices $c_{x(c),ij}$ of the three semi-local terms ($E_{x,sr}^{\text{SL}}, E_{c,ss}^{\text{SL}}, E_{c,os}^{\text{SL}}$) and $c_{x,sr}^{\text{HF}}$ into a single vector. The vector length is $3(M+1)(N+1)+1$.
- $F_{k,l}$ is the contribution of the l -th basis function to the energy for data point k , constructed from Equation 47 for system k .
- $E_k^{\text{To-fit}}$ is the energy term to fit. When using $\omega\text{B97X-V}$ as the base functional, it is defined as

$$\begin{aligned} E_k^{\text{To-fit}} &= E_k^{\text{ref}} - E_k^{\omega\text{B97X-V}} \\ &\quad + (E_{x,sr}^{\text{SL},\omega\text{B97X-V}}(\omega_0) + c_{x,sr}^{\text{HF},\omega\text{B97X-V}} E_{x,sr}^{\text{HF}}(\omega_0) + E_{c,ss}^{\text{SL},\omega\text{B97X-V}} + E_{c,os}^{\text{SL},\omega\text{B97X-V}}) \\ &\quad + E_{x,lr,k}^{\text{HF}}(\omega_0) - E_{x,lr,k}^{\text{HF}}(\omega), \end{aligned} \quad (49)$$

for system k . Therefore, changing ω from ω_0 to a trial value requires the recomputation of the corresponding long-range HF, short-range HF and semi-local exchange contributions in $F_{k,l}$.

After applying the weight, we reduce the optimization problem to a linear least-squares form,

$$L = \frac{1}{2} \|\mathbf{A}\mathbf{c} - \mathbf{b}\|^2, \quad (50)$$

where:

- $\mathbf{A}$ is a $K \times L$ matrix with elements $A_{k,l} = \sqrt{w_k} F_{k,l}$.
- $\mathbf{c}$ is a vector of the unknown coefficients c_l to be determined.
- $\mathbf{b}$ is a vector with elements $b_k = \sqrt{w_k} E_k^{\text{To-fit}}$.

This setup allows efficient determination of the coefficients c_l through standard linear least-squares techniques. However, the introduction of a base functional while necessary to create this linear least square problem, makes our fitting problem not self-consistent and the fitting results may be sensitive to the choice of base functional. To address this issue, we decided to use an existing functional with good accuracy and transferability, specifically $\omega\text{B97X-V}$, as the base functional.

3.2 Best Subset Selection

In our approach, we set $M = 11$ and $N = 7$ in Eqn. 46, resulting in $3(M+1)(N+1)+1 = 289$ linear parameters in total. Including such a high number of parameters risks overfitting, reducing the functional's ability to generalize to systems outside the training set. Best subset selection (BSS) is a statistical method used to identify the most

significant subset of variables (in this case, expansion coefficients or basis functions) that contribute meaningfully to the model’s performance, which keeps the model as simple as possible. Given a subset size s , the BSS optimization problem is formulated as

$$\min_{\mathbf{c}} \|\mathbf{A}\mathbf{c} - \mathbf{b}\|^2 \quad \text{subject to} \quad \|\mathbf{c}\|_0 \leq s, \quad (51)$$

where the ℓ_0 (pseudo)norm of a vector $\mathbf{c}$ counts the number of non-zero entries in $\mathbf{c}$, given by $\|\mathbf{c}\|_0 = \sum_{l=1}^L 1(c_l \neq 0)$, with $1(\cdot)$ denoting the indicator function. Due to the cardinality constraint, problem (Eqn. 51) is NP-hard [37]. A brute-force search through the formal combinatorial subset space would require $2^{288} \approx 10^{86}$ linear fittings, an infeasible computational task.

To address this challenge, ω B97M-V employed a forward stepwise selection strategy, a greedy algorithm that incrementally builds the model by adding variables one at a time according to predefined criteria. To improve robustness, over 10^{10} candidate subsets were explored. However, this strategy remains inefficient for our purposes. In particular, ω B97M-V performed only a single BSS optimization overall, whereas in the present work we aim to perform one BSS optimization for each functional design candidate in order to systematically assess their effects. Therefore, we adopt the mixed-integer optimization (MIO) algorithm [38], a state-of-the-art approach introduced in 2016.

Mixed-integer optimization reformulates the subset selection problem as an optimization with both continuous and integer variables. The optimization problem can be expressed as:

$$\begin{aligned} Z_1 = \min_{\mathbf{c}, \mathbf{z}} \quad & \frac{1}{2} \|\mathbf{A}\mathbf{c} - \mathbf{b}\|^2 \\ \text{s.t.} \quad & \mathbf{Q}\mathbf{c} \leq \mathbf{t}, \\ & -\mathcal{M}_U z_l \leq c_l \leq \mathcal{M}_U z_l, \quad l = 1, \dots, L, \\ & z_l \in \{0, 1\}, \quad l = 1, \dots, L, \\ & \sum_{l=1}^L z_l \leq s, \end{aligned} \quad (52)$$

where:

- $z_l \in \{0, 1\}$ is a binary variable indicating whether the coefficient c_l is included ($z_l = 1$) or excluded ($z_l = 0$) from the model.
- $\sum_{l=1}^L z_l \leq s$ imposes a sparsity constraint, limiting the maximum number of non-zero coefficients.
- $\mathcal{M}_U$ is the upper bound for the coefficients (set to 25, as in ω B97M-V).
- $\mathbf{Q}\mathbf{c} \leq \mathbf{t}$ represents physical constraints, which will be described in detail in the next subsection.

We implement the MIO formulation using the Gurobi solver [39], with a wall-time limit of 1–2 hours per value of s , depending on the subset size, and one restart. All calculations are performed using 16 CPU cores. We scan subset sizes from $s = 24$ to

80, resulting in a total computational cost of approximately 3000 CPU hours for a single functional design candidate.

By employing MIO, we aim to achieve a more efficient and closer-to-global solution of the subset selection problem, thereby improving both the performance and the generalizability of the resulting functional.

3.3 Numerical Constraints

Using F_{corr} alone to satisfy physical constraints can be sufficient, but it may introduce unnecessary rigidity and limit flexibility. Instead, certain constraints can be enforced numerically on a mesh grid defined over $u_{x\sigma(\text{cos},\text{css})}$ and β_σ (or equivalently w_σ). This strategy ensures that the functional satisfies specific physical constraints while preserving flexibility for overall performance optimization.

For constraints such as (1) and (7), we construct a dense mesh grid that spans the full variable domain of β (or w) and u . Specifically, we choose $2\beta - 1$ (or w) values as $[-1.0, -0.7, -0.2, -0.08, -0.01, 0.01, 0.05, 0.1, 0.3, 0.5, 0.65, 0.7, 0.8, 0.95, 1.0]$ and u values as $[0.001, 0.01, 0.1, 0.15, 0.22, 0.3, 0.6, 0.7, 0.85, 0.95, 0.99]$. We then enforce these constraints as inequality conditions on the ICFs over the grid. The smooth expansion functions therefore satisfy the constraints over most regions of the variable domain, although not strictly everywhere.

For constraints such as (6), we enforce the condition along a one-dimensional line in variable space rather than over a two-dimensional grid. If β is used, the line corresponding to a one-electron density is simply given by $\beta = 0$, and we use u values in the range $[0.001, 0.01, 0.1, 0.15, 0.22, 0.3, 0.5, 0.6, 0.7, 0.85, 0.95, 0.99, 0.999]$. If w_σ is used, the procedure is slightly more complex. We sample s^2 values across $[0.3, 0.6, 1.2, 1.8, 2.4, 2.7, 3.0, 3.3, 3.6, 3.9, 4.2, 4.5, 4.8, 5.2, 5.6, 6.0, 6.6, 7.2, 7.8, 8.4, 9.0, 10.2, 11.4, 12.0, 13.2, 15.0, 18, 24, 36, 60, 120]$ and compute the corresponding w and u values.

Although numerical enforcement provides greater flexibility, it has two notable limitations. First, a finite mesh cannot guarantee strict satisfaction of constraints over the entire continuous variable space, particularly in edge cases such as $u \rightarrow 1^-$. However, with a sufficiently dense mesh, any residual violations are confined to very small regions and are unlikely to materially affect practical accuracy, as shown in Section 6.1. Second, this approach is most suitable for inequality constraints, because enforcing equality constraints on a mesh would rigidly fix values at many grid points and thereby reduce the number of effective nonzero coefficients available for fitting. This would consume degrees of freedom that would otherwise be used to optimize overall performance.

Beyond physical constraints, we also introduce grid-sensitivity constraints to improve numerical stability across different integral quadrature resolutions. Specifically, we construct the matrix $\mathbf{A}$ in Equation 52 using the high-resolution grid (250,974)/SG-1. In parallel, we construct $\mathbf{A}'$ using the lower-resolution grid (99,590)/SG-1. Because enforcing inequality constraints over all data points is computationally expensive, we adopt a selective strategy. We first optimize the functional without grid-sensitivity constraints to identify promising candidates. For each candidate, we then collect the 100 data points with the largest deviations $|(\mathbf{A}_{k,:} - \mathbf{A}'_{k,:})\mathbf{c}|$ and an additional 200 data points with the largest ℓ_1 -norms of $\mathbf{A}_{k,:} - \mathbf{A}'_{k,:}$. The union of these data points is used to enforce grid-sensitivity constraints. For these selected data points, the energy

differences between the two grids are required to remain below a predefined threshold, set to 0.015 kcal/mol in most cases (except in the exploration of Section 4.2). This procedure ensures that the functional remains stable and reliable across varying grid resolutions while maintaining computational efficiency.

3.4 Gold-Standard Chemical Database

In this work, we use the Gold-Standard Chemical Database (GSCDB) [40, 41] as the source of training and validation data. Because database development and functional development proceeded in parallel over more than two years, multiple versions of GSCDB were used at different stages of the project. Unless otherwise noted, most training cycles employ the version used in Jiashu Liang’s thesis work [40] (GSCDB138), whereas the final assessment uses the version reported in the published paper [41] (GSCDB137).

Functional accuracy on GSCDB137 is assessed using the error definitions adopted in Ref. 41, with dataset-specific conventions where appropriate. For most data sets, we compute the mean absolute error (MAE). For Pol132, HR46, T144, OEEF, and OEEFD, we instead use the mean absolute relative error (MARE) to account for the wide variation in the magnitudes of the reference values. For Dip146, we use the mean absolute regularized error defined in the original study [42]. For TMD10, MOR13, and TMB11, we follow the recommendations of the original publications and adopt the corresponding estimated (weighted) mean absolute deviations. For O24 and O24x4, we apply additional data-point weights based on the typical performance of hybrid functionals on each case within O24. All weighting factors are provided on GitHub at: <https://github.com/JiashuLiang/GSCDB>

To compare functionals consistently across chemically diverse data sets, we quantify performance using the normalized error ratio (NER) defined in Ref. 41. For each data set, the NER is defined as the ratio between the functional’s error and a data-set-specific “standard error.” In the final assessment, we use the same standard errors as reported in Ref. 41, namely the average of the second-, third-, and fourth-lowest errors among all 14 tested hybrid functionals. Hybrid functionals are used as the reference class because of their widespread use in chemistry, and excluding the best-performing functional helps reduce bias from potential accidental overfitting on individual data sets.

In earlier cycles, because the benchmark survey was not yet complete, we defined the standard error for each data set as the average of the two lowest root-mean-squared errors (RMSEs) obtained among four reference functionals: ω B97X-V, ω B97M-V, r2SCAN0-D4, and CF22D.

The resulting NER values are then averaged over all data sets within each property category—barrier heights (BH), electric field responses (EF), vibrational frequencies (FREQ), isomerization energies (ISO), noncovalent interactions (NC), intramolecular noncovalent interactions (INC), thermochemical properties (TC), and transition metal systems (TM)—and also over all data sets in the full GSCDB to yield the overall mean score used for each functional candidate.

3.5 Training cycles, data selection, and weighting

Our optimization is organized into **training cycles**. Each cycle consists of: (i) generating Kohn-Sham orbitals and energies with a chosen base functional; (ii) selecting data-set inclusion and determining weights for that cycle; (iii) performing exploration to choose the design options and constraints; and (iv) training to identify the best functional candidate under the chosen design. The resulting best candidate is then used as the base functional for the next cycle. In total, we performed two training cycles. We stopped when the self-consistent energy is nearly the same as the energy fitted with a chosen base functional on most data sets.

The weights w_k in the loss function allow us to prioritize particular data points and property classes. For example, especially important properties, or data points with smaller numerical magnitudes, can be assigned larger weights. This flexibility enables us to tailor the fitting process to achieve strong overall performance for the intended applications of the functional.

We used an active-learning strategy similar to that used for CF22D [43] to determine data-set inclusion and weights during development, starting from the weights used to train ω B97M-V. For the final cycle, the inclusion list and weights were fixed before the best-subset optimization. Table 2 reports the final weights and the number of fitting entries actually implied by the recorded selection. Rows marked **A11** in the machine-readable record contain every reaction in that data set; otherwise, only the explicitly listed reaction identifiers are included. For AE18, **Shrink** assigns the j th listed atom the weight $w_j = 1/\sqrt{j}$; every other tabulated weight is a constant w_k multiplying that entry’s squared residual in Eq. 43.

Table 2: Final coefficient-fitting data and weights. The property classes use the manuscript groupings BH (barrier heights), EF (electric-field response properties), ISO (isomerization energies), INC (intramolecular noncovalent interactions), NC (intermolecular noncovalent interactions), TC (thermochemistry), and TM (transition-metal chemistry). The selection column gives either all reactions in the data set or the complete ordered reaction list; the # column gives the corresponding number of fitting entries.

Set or subset	Class	#	w_k	Included reactions
BH28	BH	28	10	A11
BHDIV7	BH	7	16	A11
BHPERI11	BH	11	30	A11
CRBH14	BH	14	10	A11
DBH22	BH	22	30	A11
PX9	BH	9	20	A11

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Set or subset	Class	#	w_k	Included reactions
Dip146	EF	63	1600	Dip146.1, Dip146.4, Dip146.7, Dip146.10, Dip146.13, Dip146.16, Dip146.19, Dip146.22, Dip146.25, Dip146.28, Dip146.32, Dip146.35, Dip146.38, Dip146.41, Dip146.44, Dip146.47, Dip146.50, Dip146.53, Dip146.56, Dip146.59, Dip146.62, Dip146.65, Dip146.68, Dip146.71, Dip146.74, Dip146.77, Dip146.80, Dip146.83, Dip146.86, Dip146.89, Dip146.92, Dip146.95, Dip146.98, Dip146.101, Dip146.104, Dip146.107, Dip146.110, Dip146.113, Dip146.116, Dip146.119, Dip146.122, Dip146.125, Dip146.128, Dip146.131, Dip146.134, Dip146.137, Dip146.140, Dip146.143, Dip146.146, Dip146.149, Dip146.152, Dip146.155, Dip146.158, Dip146.161, Dip146.164, Dip146.167, Dip146.170, Dip146.173, Dip146.176, Dip146.179, Dip146.182, Dip146.185, Dip146.188
HR46	EF	45	10000	HR46.1, HR46.4, HR46.5, HR46.8, HR46.10, HR46.12, HR46.13, HR46.17, HR46.20, HR46.23, HR46.28, HR46.30, HR46.34, HR46.37, HR46.38, HR46.42, HR46.45, HR46.48, HR46.51, HR46.53, HR46.57, HR46.59, HR46.62, HR46.66, HR46.70, HR46.71, HR46.75, HR46.76, HR46.80, HR46.82, HR46.86, HR46.87, HR46.91, HR46.94, HR46.97, HR46.99, HR46.104, HR46.105, HR46.110, HR46.113, HR46.114, HR46.119, HR46.120, HR46.123, HR46.126
Pol130	EF	95	6400	Pol130.1, Pol130.4, Pol130.11, Pol130.12, Pol130.16, Pol130.20, Pol130.22, Pol130.24, Pol130.26, Pol130.28, Pol130.33, Pol130.36, Pol130.39, Pol130.42, Pol130.44, Pol130.48, Pol130.51, Pol130.54, Pol130.56, Pol130.59, Pol130.62, Pol130.66, Pol130.68, Pol130.70, Pol130.74, Pol130.81, Pol130.85, Pol130.91, Pol130.95, Pol130.98, Pol130.101, Pol130.102, Pol130.105, Pol130.107, Pol130.111, Pol130.115, Pol130.116, Pol130.121, Pol130.123, Pol130.126, Pol130.128, Pol130.130, Pol130.132, Pol130.137, Pol130.142, Pol130.144, Pol130.145, Pol130.147, Pol130.149, Pol130.153, Pol130.157, Pol130.158, Pol130.161, Pol130.165, Pol130.167, Pol130.170, Pol130.171, Pol130.173, Pol130.176, Pol130.180, Pol130.182, Pol130.185, Pol130.188, Pol130.192, Pol130.197, Pol130.200, Pol130.203, Pol130.204, Pol130.209, Pol130.211, Pol130.214, Pol130.217, Pol130.223, Pol130.227, Pol130.229, Pol130.231, Pol130.234, Pol130.237, Pol130.242, Pol130.245, Pol130.248, Pol130.250, Pol130.254, Pol130.257, Pol130.261, Pol130.262, Pol130.267, Pol130.273, Pol130.276, Pol130.279, Pol130.282, Pol130.284, Pol130.290, Pol130.294, Pol130.296
EIE22	ISO	22	20	All
PCONF21	INC	18	50	All
Styrene42	ISO	42	5	All
AlkIsomer11	ISO	11	50	All
BUT14DIOL	INC	14	200	BUT14DIOL.3, BUT14DIOL.4, BUT14DIOL.9, BUT14DIOL.14, BUT14DIOL.17, BUT14DIOL.25, BUT14DIOL.37, BUT14DIOL.38, BUT14DIOL.40, BUT14DIOL.43, BUT14DIOL.46, BUT14DIOL.47, BUT14DIOL.58, BUT14DIOL.60
S66Rel7	NC	74	60	S66Rel7.1, S66Rel7.8, S66Rel7.15, S66Rel7.22, S66Rel7.29, S66Rel7.36, S66Rel7.43, S66Rel7.50, S66Rel7.57, S66Rel7.64, S66Rel7.71, S66Rel7.78, S66Rel7.85, S66Rel7.92, S66Rel7.97, S66Rel7.99, S66Rel7.113, S66Rel7.118, S66Rel7.120, S66Rel7.127, S66Rel7.134, S66Rel7.139, S66Rel7.141, S66Rel7.146, S66Rel7.148, S66Rel7.153, S66Rel7.155, S66Rel7.160, S66Rel7.162, S66Rel7.167, S66Rel7.169, S66Rel7.176, S66Rel7.183, S66Rel7.188, S66Rel7.190, S66Rel7.197, S66Rel7.204, S66Rel7.209, S66Rel7.211, S66Rel7.218, S66Rel7.223, S66Rel7.225, S66Rel7.230, S66Rel7.232, S66Rel7.239, S66Rel7.246, S66Rel7.253, S66Rel7.267, S66Rel7.274, S66Rel7.279, S66Rel7.281, S66Rel7.288, S66Rel7.295, S66Rel7.309, S66Rel7.316, S66Rel7.323, S66Rel7.330, S66Rel7.337, S66Rel7.344, S66Rel7.351, S66Rel7.358, S66Rel7.365, S66Rel7.372, S66Rel7.377, S66Rel7.379, S66Rel7.393, S66Rel7.405, S66Rel7.407, S66Rel7.414, S66Rel7.419, S66Rel7.421, S66Rel7.435, S66Rel7.442, S66Rel7.449
Bauza30	NC	30	20	All
CT20	NC	20	400	All
HCP32	NC	32	60	All
TA13	NC	13	8	All
XB25	NC	8	200	XB25.1, XB25.2, XB25.3, XB25.4, XB25.5, XB25.6, XB25.7, XB25.8
HW6CI5	NC	5	120	All
Shields38	NC	8	100	Shields38.14, Shields38.15, Shields38.16, Shields38.17, Shields38.18, Shields38.19, Shields38.20, Shields38.21

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Set or subset	Class	#	w_k	Included reactions
A24	NC	24	400	A11
DS14	NC	14	60	A11
HSG	NC	21	150	A11
IHB100	NC	100	80	A11
IL16	NC	16	120	A11
RG10N1	NC	23	25	RG10N.1, RG10N.8, RG10N.31, RG10N.33, RG10N.58, RG10N.59, RG10N.61, RG10N.63, RG10N.92, RG10N.93, RG10N.95, RG10N.125, RG10N.147, RG10N.149, RG10N.176, RG10N.177, RG10N.179, RG10N.181, RG10N.210, RG10N.212, RG10N.240, RG10N.242, RG10N.244
RG10N2	NC	17	800	RG10N.3, RG10N.9, RG10N.11, RG10N.40, RG10N.71, RG10N.102, RG10N.105, RG10N.128, RG10N.131, RG10N.151, RG10N.153, RG10N.184, RG10N.189, RG10N.216, RG10N.218, RG10N.249, RG10N.253
S22	NC	22	120	A11
MB16-43	TC	43	0.05	A11
MX34	TC	34	0.02	A11
AE18	TC	18	$1/\sqrt{j}$	A11
ALK8	TC	8	2	A11
AlkAtom19	TC	19	1	A11
BDE99nonMR	TC	83	1.2	A11
BSR36	TC	36	8	A11
CR20	TC	20	15	A11
DIPCS9	TC	9	0.6	A11
G21EA	TC	25	9	A11
G21IP	TC	36	1.4	A11
HEAVYSB11	TC	11	0.8	A11
RSE43	TC	43	2	A11
TAE.W4-17nonMR	TC	183	0.8	A11
3d4dIPSS	TM	32	1.5	A11
CUAGAU83	TM	36	1.5	CUAGAU.1, CUAGAU.2, CUAGAU.3, CUAGAU.4, CUAGAU.5, CUAGAU.8, CUAGAU.9, CUAGAU.10, CUAGAU.11, CUAGAU.12, CUAGAU.13, CUAGAU.14, CUAGAU.15, CUAGAU.16, CUAGAU.17, CUAGAU.18, CUAGAU.19, CUAGAU.22, CUAGAU.23, CUAGAU.24, CUAGAU.25, CUAGAU.26, CUAGAU.27, CUAGAU.28, CUAGAU.29, CUAGAU.30, CUAGAU.31, CUAGAU.32, CUAGAU.33, CUAGAU.36, CUAGAU.37, CUAGAU.38, CUAGAU.39, CUAGAU.40, CUAGAU.41, CUAGAU.42
MOR13	TM	13	10	A11
TMB11	TM	11	10	A11
TMD10	TM	10	10	A11

4 Design exploration

4.1 Effect of functional design choices

We investigated several design choices in constructing the new functional. We would like to quantify how each choice affects performance, particularly when the choices are motivated by physical constraints.

We first considered six two-level factors (i.e., binary yes/no choices):

- **Exchange NUSC (X NUSC):** Whether to apply SCAN’s non-uniform scaling correction (NUSC) factor in the exchange functional to fully satisfy constraint (5).
- **SCAN correlation base (SCAN base):** Whether to use SCAN ($\alpha = 1$, UEG case) rather than PW92 as the base functional for correlation to satisfy constraints (9) and (12).
- **Same-spin SCC (Css SCC):** Whether to employ the proposed self-correlation correction (SCC) factor ($2\beta_\sigma$) in the same-spin correlation to satisfy constraint (11).
- **Exchange β variable (X β):** Whether to use β rather than w as the τ -related dimensionless variable for the exchange fitting factor.
- **Same-spin β variable (Css β):** Whether to use β rather than w as the τ -related dimensionless variable for the same-spin correlation fitting factor.
- **Opposite-spin β variable (Cos β):** Whether to use β rather than w as the τ -related dimensionless variable for the opposite-spin correlation fitting factor.

In addition, we explored six three-level factors (i.e. three-way choices) corresponding to the choice of polynomial family—Linear, Legendre (Leg), or Chebyshev (Che)—used to expand the functional components. These factors cover the polynomial choices for the u variables in X, Css, and Cos, as well as for the w (or β) variables in X, Css, and Cos. Their outcomes are discussed in Section 4.3.

To efficiently evaluate multiple factors simultaneously, we employed an orthogonal experimental design. Orthogonal arrays reduce the number of experiments required while maintaining balanced sampling of factor levels. We used an L_{36} orthogonal array, which is the smallest design that covers all factors considered here. Consistent with the subset-selection procedure described in Section 3.2, we scanned the subset size s from 24 to 80, yielding 57 functional candidates for each of the 36 experiments. As in our fixed-orbital screening protocol, we assessed functional performance using the normalized error ratio (NER) defined in Section 3.4. Figure 2 summarizes the average change in the five smallest values of this metric when adopting the alternative (new) design choice relative to the ω B97M-V choice for each two-level factor. Negative changes indicate improvement.

Although a strict statistical analysis would likely classify all effects as insignificant due to their small magnitudes, we adopted an empirical significance threshold of 0.02 to highlight the large effects. Using this criterion, we draw the following conclusions:

1. **X NUSC:** The effect is insignificant.
2. **SCAN base:** Using SCAN as the correlation base generally worsens performance.
3. **Css SCC:** Introducing the SCC factor $2\beta_\sigma$ slightly improves performance for EF properties while slightly worsening performance for TM systems.
4. **X β :** Using β in exchange generally improves performance, except in the TC category.
5. **Css β :** Using β in same-spin correlation improves performance for NC, TC, and TM systems but is detrimental for FREQ.
6. **Cos β :** Using β in opposite-spin correlation substantially worsens most categories.

Our strategy is to satisfy as many physical constraints as possible, and we therefore adopt the more physically reasonable design choice provided it does not substantially degrade performance. Accordingly, we adopt the exchange NUSC factor and use β for the exchange and same-spin correlation fitting factors.

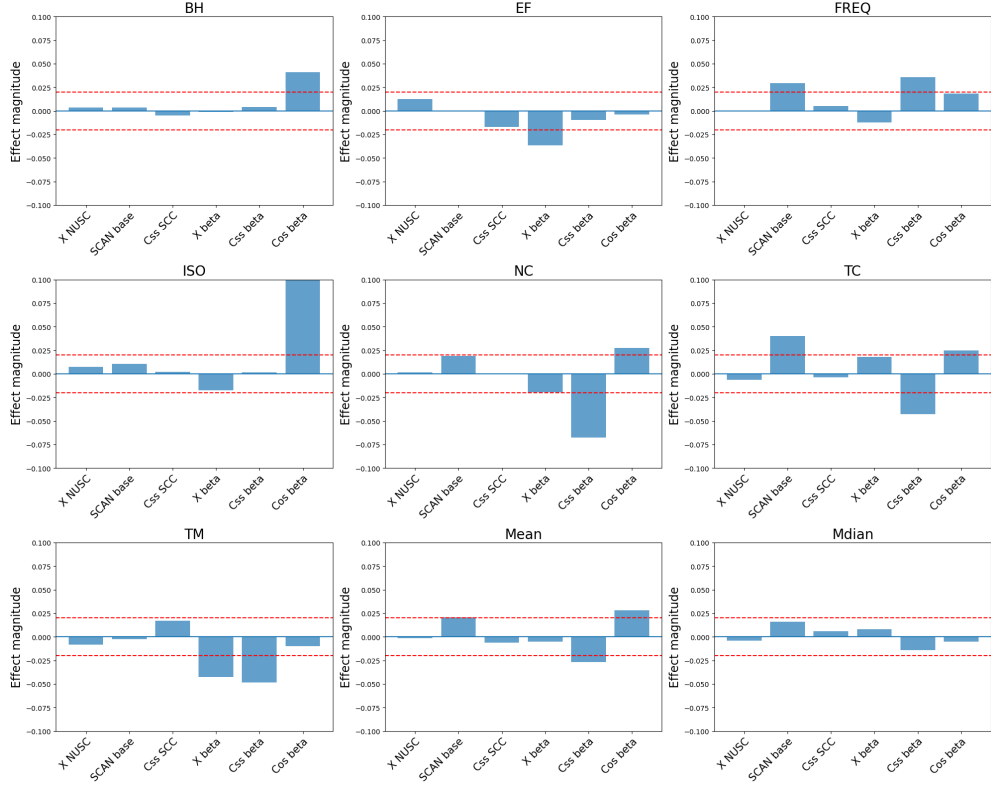


Fig. 2 Effects of the two-level factors from orthogonal design experiments using an L_{36} array. For each functional candidate, we compute the NERs and average within each property category (e.g., BH, EF) and over the full database. The plotted metric is the average change in the five smallest values of this ratio under each factor change. Bars report the change when moving from the ω B97M-V design choice to the alternative choice for each factor; negative values indicate improvement if adopting new design. Red dashed lines denote the empirical significance threshold of 0.02.

Because orthogonal designs can introduce incidental coupling among factors and may not perfectly reflect outcomes in the fully optimized setting, we then performed full factorial experiments to further evaluate the remaining three factors—SCAN base, Ccs SCC, and Cos β . We enumerated all combinations of these factors ($2^3 = 8$), scanned the subset size s from 24 to 80 for each setup, and compared performance using the same NER-based metric. The results are summarized in Figure 3.

From the overall median, all three factors lead to slight performance degradation, although the effects are not significant. In contrast, the overall mean indicates that only using β as the τ -related dimensionless variable in opposite-spin correlation (**Cos** β) substantially worsens performance relative to using w . This trend is physically plausible: the Ccs SCC factor and the SCAN correlation base are both introduced to satisfy specific physical constraints, whereas β is most closely associated with one-electron orbital characteristics and is therefore less meaningful for opposite-spin correlation. By

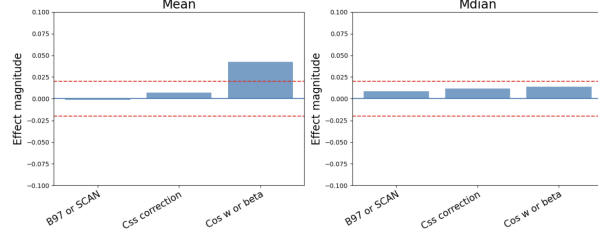


Fig. 3 Effects of the remaining two-level factors from full factorial experiments. The metric is computed using the same procedure as in Figure 2, based on the average change in the five smallest values of the NER-based ratio under each factor combination.

comparison, β plays a direct role in enforcing constraints (6) and (11) for exchange and same-spin correlation, respectively. Accordingly, the numerical-stability advantages of the w variable remain preferable for opposite-spin correlation. Based on these results, we adopt SCAN as the correlation base functional and include the same-spin SCC factor, but we do not use β for the opposite-spin correlation fitting factor.

4.2 Effect of numerical constraints

In the previous subsection, we focused on satisfying selected physical constraints through the analytical design of the functional form, including different choices of F_{corr} . Here, we examine the effect of enforcing additional constraints numerically on the mesh grids described in Section 3.3. Specifically, we consider five constraints enforced numerically: (1) exchange negativity, (4a) the uniform-electron-gas (UEG) exchange limit, (6) the tight bound for spin-unpolarized two-electron densities, (7) non-positivity of correlation, and (14) the tighter general Lieb–Oxford (Chan–Handy) bound. In addition, we investigate how varying the grid-sensitivity threshold influences functional performance.

Figure 4 summarizes the effect of different enforced physical-constraint sets. The column labels correspond to the constraint indices defined in Section 2, and the row labels denote different data-set categories. The first row, labeled OF, represents overfitting and is defined using the average NER of the AE11, MB08-165, and MB16-43 data sets. We selected these three data sets because we consider them the least similar to the training data and therefore useful for assessing overfitting. As expected, the OF metric is strongly affected by constraint enforcement, indicating that physical constraints can substantially reduce overfitting. By contrast, the inclusion or exclusion of individual constraints has only a minor impact on overall performance, as reflected by similar mean and median NER values, with the exception of the first column.

A notable exception arises when comparing the first two columns, which differ only by enforcement of constraint (7), non-positivity of correlation: enforcing (7) increases the overall mean NER by 0.02. This behavior is consistent with reduced exchange–correlation error cancellation, for example when overly negative exchange is compensated by spuriously positive correlation. While enforcing (7) improves EF performance, it substantially worsens ISO and NC accuracy. Therefore, consistent with

our goal of enforcing as many constraints as possible without materially compromising performance, we retain constraints (1), (4a), (6), and (14), but do not enforce (7).

	1,4a,6,7,14	1,4a,6,14	1,4a,14	1,14	1
OF	1.092	1.043	1.066	1.372	1.303
BH	0.968	0.953	0.957	0.955	0.964
EF	1.056	1.193	1.163	1.107	1.108
FREQ	0.885	0.880	0.883	0.897	0.925
ISO	1.504	1.434	1.447	1.340	1.347
NC	1.178	1.125	1.167	1.152	1.149
TC	1.003	0.988	1.036	1.058	1.051
TM	1.203	1.237	1.214	1.246	1.230
Mean	1.069	1.049	1.063	1.052	1.048
Median	0.962	0.959	0.958	0.950	0.954

Fig. 4 Effect of enforcing different numerical physical constraints on functional performance, as measured by the NER (i.e. smaller is better). Column labels correspond to the constraint indices defined in Section 2. For each functional candidate, NERs are averaged within each property category (e.g., BH, EF) and over the full database. The row labeled OF represents overfitting, defined using the average of the NERs on the AE11, MB08-165, and MB16-43 data sets.

Figure 5 summarizes the impact of different grid-sensitivity thresholds on overall performance. As described in Section 3.3, we adopt a two-cycle procedure. In the first cycle, no grid-sensitivity constraints are applied. In the second cycle, grid constraints are enforced on 452 data points that exhibit the largest deviations in the first cycle. This two-cycle protocol is the default setting used in the other subsections and corresponds to the first column in Figure 5.

Even under this default setting, 10 data points still show grid deviations exceeding 0.015 kcal/mol for the optimized functional. To strengthen grid-stability control in the present analysis, we used both the first- and second-cycle optimized functionals to identify a larger set of 609 data points, and enforced grid constraints on these 609 points for all subsequent optimizations in this subsection.

The first row of Figure 5 reports the average number of non-zero coefficients in the best five functionals (defined as those with the smallest median metric). Applying moderate grid constraints (thresholds of 0.06 or 0.10 kcal/mol) does not materially change the number of non-zero coefficients, indicating that flexibility is largely retained, while nevertheless reducing grid sensitivity. Tightening the threshold from 0.06 to 0.03 kcal/mol increases the number of non-zero coefficients by six and reduces the number of data points with large grid deviations (99590# > 0.015 and 75302# > 0.1) by approximately 50%. Further tightening to 0.015 kcal/mol increases the number of non-zero coefficients by eight, but yields only a modest additional improvement in grid stability.

Across property categories, EF, FREQ, ISO, and TM benefit from tighter grid constraints, whereas OF, BH, NC, and TC show slight performance declines. Although the strictest threshold of 0.015 kcal/mol leads to a small overall performance penalty relative to looser thresholds, we retain it because of the substantial reduction in EF

	0.015*	0.015	0.03	0.06	0.10	None
#nz coefficients	73.0	74.4	66.4	60.8	60.8	61.2
99590 max diff	0.022	0.015	0.030	0.060	0.100	11.480
75302 max diff	21.424	30.338	21.283	16.074	11.670	51.254
99590 #>0.015	10	0	113	284	401	1860
99590 #>0.1	0	0	0	0	0	208
99590 #>0.5	0	0	0	0	0	18
75302 #>0.015	1556	1509	2018	2452	2693	3787
75302 #>0.1	100	103	122	205	259	820
75302 #>0.5	7	6	6	9	12	92
OF	1.027	1.043	1.098	0.827	0.815	0.808
BH	0.951	0.953	0.908	0.882	0.869	0.881
EF	1.196	1.193	1.600	1.880	1.914	2.176
FREQ	0.900	0.880	0.928	0.952	0.905	0.985
ISO	1.343	1.434	1.392	1.435	1.636	1.508
NC	1.097	1.125	1.110	1.081	1.100	1.054
TC	0.978	0.988	0.990	0.970	0.925	0.956
TM	1.240	1.237	1.367	1.388	1.356	1.358
Mean	1.029	1.049	1.041	1.044	1.045	1.047
Median	0.950	0.959	0.920	0.923	0.909	0.918

Fig. 5 Effect of grid-sensitivity constraints on functional performance. Values in the bottom half of the figure report the average NER metric of the best five functional candidates across the 8 data categories, as well the mean and median. The column labeled “0.015*” denotes the default setting used in the other subsections, whereas subsequent columns correspond to enforcing grid sensitivity constraints on larger sets of data points. Row labels such as 99590# > 0.015 indicate the number of data points for which the energy difference between the (99,590) grid and the reference fine grid (250,974) exceeds 0.015 kcal/mol.

errors and its importance for practical applications. Finally, comparing the 0.015 kcal/mol threshold under two enforcement strategies (fewer versus more constrained points) shows that the default setting with fewer constrained points provides slightly better performance with comparable grid stability. Accordingly, we retain this default grid-sensitivity protocol in the final training procedure.

4.3 Effect of polynomial expansions choices in the fitting factors

In the L_{36} orthogonal experiment described in Section 4.1, we also explored six three-level factors corresponding to the choice of polynomial family—Linear (Lin), Legendre (Leg), or Chebyshev (Che)—used to expand the fitting factor of functional components (X, C_{ss}, C_{os}). We compared Legendre and Chebyshev polynomials against monomial basis for each component, as shown in Figure 6. If both Legendre and Chebyshev yield positive changes, Linear is preferred; otherwise, the more negative change indicates the preferred polynomial family.

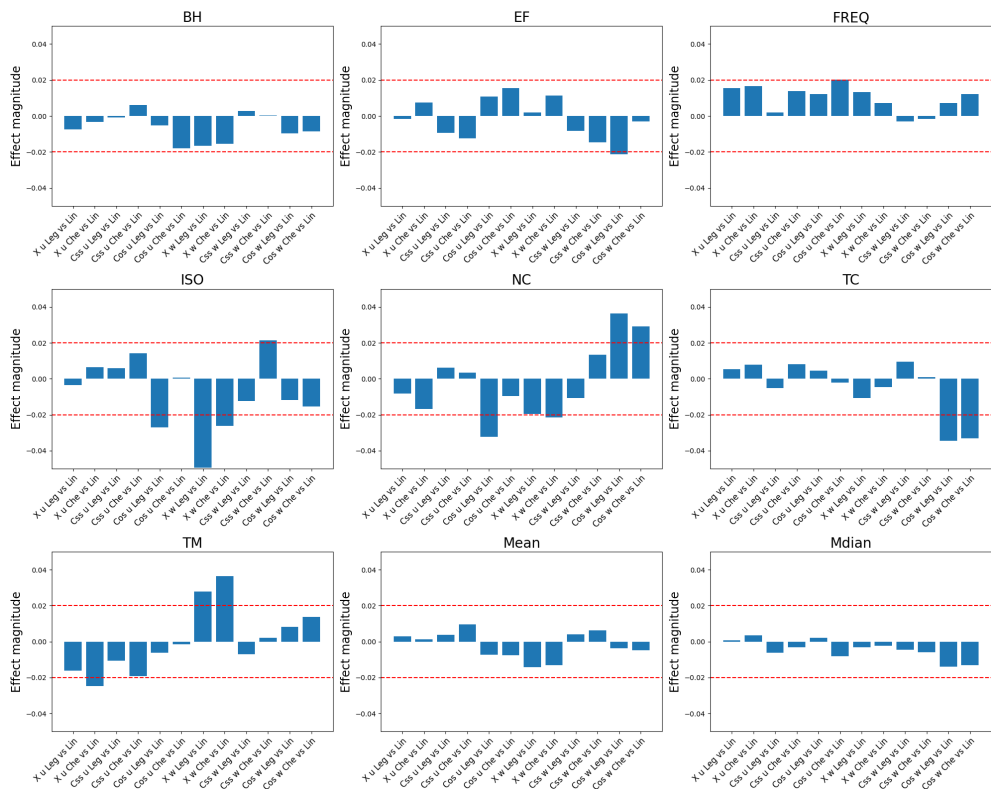


Fig. 6 Comparison of polynomial families (Linear, Legendre, and Chebyshev) used for selected expansion components (X, Csa, Cos). Bars show the change in the NER metric for Legendre (Leg) and Chebyshev (Che) relative to Linear (Lin). Positive values indicate worse performance than Linear, whereas a more negative value indicates a preferred alternative. Red dashed lines denote the empirical significance threshold of 0.02.

From the overall mean and median across all data sets, the polynomial choice is not significant for any component. However, for specific property types, the effect can be substantial. Therefore, we initially selected between Linear and Legendre polynomials for each component based on the overall mean performance, and we reevaluated these choices in the final training cycle.

In the final cycle, we performed an additional screening in which we replaced the expansion for one component (X, Csa, or Cos) at a time. For each setting, we scanned subset sizes $s = 22$ to 89, yielding 68 candidate functionals, and computed the average number of non-zero coefficients among candidates that achieve an overall mean $\text{NER} \leq 0.95$. This metric serves as a practical proxy for how efficiently a given expansion choice reaches the target performance: fewer non-zero coefficients correspond to fewer fitted degrees of freedom (DoF), and lower DoF is expected to reduce the probability of overfitting, with potential benefits for subsequent applications to independent chemical problems.

	Exchange			Same-spin Correlation			Opposite-spin Correlation		
	u Lin	u Leg	u Cheb	u Lin	u Leg	u Cheb	u Lin	u Leg	u Cheb
β (or w) Lin	65.00	72.33	62.33	72.00	65.67	69.67	75.00	82.33	82.33
β (or w) Leg	72.00	85.00	76.67	59.00	60.33	65.00	68.33	72.00	71.67
β (or w) Cheb	72.67	76.00	74.00	60.67	71.67	61.33	78.67	72.67	69.00

Fig. 7 Average number of non-zero coefficients among the candidates (68 per setting, from scanning subset sizes $s = 22$ to 89) that achieve an overall mean $\text{NER} \leq 0.95$. Columns correspond to the polynomial-family choice for the u expansion in exchange (X), same-spin correlation (Css), and opposite-spin correlation (Cos), each with Linear, Legendre, and Chebyshev options. Rows correspond to the polynomial-family choice for the β (or w) expansion (Linear, Legendre, Chebyshev). Lower values indicate that the target performance is reached with fewer non-zero coefficients on average.

Figure 7 indicates that no single polynomial family is uniformly optimal across components. Instead, the best-performing choices depend on both the component and the variable. Several combinations appear favorable, for example Chebyshev for u together with Linear for β in exchange, and Linear for u together with Legendre for β in same-spin correlation.

However, combining the individually best choices for X, Css, and Cos into a single functional design did not yield the expected improvement. This suggests that the polynomial-family effects are significant but not additive, and that they can interact with other design choices in a non-synergistic manner. Therefore, for the final design we adopt a conservative and robust combination: Linear for u and Legendre for β in both exchange and same-spin correlation, and Legendre for both u and w in opposite-spin correlation.

4.4 Effect of nonlinear parameters

The nonlinear parameters were examined using discrete grid searches, with the remaining nonlinear parameters held fixed at each stage. We first performed a broad screening scan and then repeated a narrower scan during the final training cycle after the functional-design choices had been fixed; the exact grids are given below.

$\omega\text{B97M-V}$ adopted its nonlinear parameter values directly from previous studies: $\gamma_x = 0.004$, $\gamma_{c,ss} = 0.2$, $\gamma_{c,os} = 0.006$, $\omega = 0.3$, and VV10 parameters $b = 6$ and $C = 0.01$. In COACH, however, we retain only γ_x , $\gamma_{c,os}$, and C . The parameters γ_x , $\gamma_{c,ss}$, and $\gamma_{c,os}$ appear primarily in the finite-domain transformations of the reduced density gradient (see Eq. 15). Because these parameters were fitted to simple atomic systems and have historically shown good transferability, we do not expect refitting to be necessary in principle. However, Matito et al. demonstrate that the historical B97 value of $\gamma_{c,ss} = 0.2$ can lead to oscillatory functional derivatives and reduced numerical stability along potential-energy surfaces [12]. Accordingly, we retrain $\gamma_{c,ss}$ to obtain a smaller value. We also retrain the range-separation parameter ω , which determines the partitioning between short-range semi-local exchange and long-range Hartree-Fock exchange, because modest changes in ω can have a large impact on functional performance. For the VV10 parameters, we keep $C = 0.01$, which was fitted to accurate asymptotic van der Waals C_6 coefficients and is expected to be transferable.

In contrast, we retrain b , since b governs the short-range damping of the VV10 kernel and should therefore be closely coupled to the newly trained semi-local part.

4.4.1 Effect of the VV10 damping parameter b

The b parameter in VV10 controls the degree of short-range damping applied to the R^{-6} dispersion asymptote, with smaller values of b decreasing the damping and thus enhancing the influence of the VV10 dispersion correction. The value of b is 5.9 in the original VV10 paper, while it is 6.0 in the ω B97X-V and ω B97M-V functionals.

Figure 8 shows the performance results for the five-point broad scan $b = \{5.0, 5.3, 5.5, 5.7, 6.0\}$, excluding any three-body dispersion contribution. For each dispersion setting, we computed the dispersion correction energies and adjusted the vector $\mathbf{b}$ in Eq. 52. We used the same metric as in the previous subsections.

	b=5.0	b=5.3	b=5.5	b=5.7	b=6.0
OF	0.995	0.903	1.041	1.039	1.024
BH	0.937	0.949	0.942	0.945	0.951
EF	1.429	1.266	1.396	1.197	1.195
FREQ	0.904	0.878	0.917	0.934	0.900
ISO	1.259	1.234	1.272	1.490	1.345
NC	1.094	1.081	1.128	1.140	1.098
TC	0.958	0.957	1.024	0.995	0.979
TM	1.270	1.232	1.241	1.218	1.239
Mean	1.015	1.005	1.039	1.053	1.030
Median	0.934	0.933	0.944	0.955	0.951

Fig. 8 Performance of the functional with different VV10 b parameters. For each functional candidate, NERs are averaged within each property category (e.g., BH, EF) and over the full database.

Based on the overall mean and median metrics, the best performance is achieved at $b = 5.3$ (1.005 and 0.933), with a notable improvement (-0.02) relative to the default b value, suggesting that $b = 5.3$ provides the best balance across all data types. The only significant exception is Electric Field Properties (EF). Because EF primarily probes the accuracy of the electronic density and VV10 has little effect on the density (as shown in Ref. 44), one possible explanation is that stronger VV10 contributions (smaller b values) make the functional more prone to overfitting other data types. Nevertheless, because the degradation is small and the overall performance improves, we decided to explore values around 5.3 in the final training cycle including three-body contributions.

In the final training cycle, we employed D4-ATM as the three-body empirical dispersion term with initial parameters $s_6 = s_8 = 0$ (to disable the two-body contribution), $s_9 = 1.0$, $a_1 = 0.4$, and $a_2 = 5.0$. Holding these initial D4-ATM parameters fixed, we separately refined b on the five-point uniform grid $b = \{5.2, 5.3, 5.4, 5.5, 5.6\}$, corresponding to a step size of 0.1. The best functional obtained corresponds to $b = 5.5$.

The D4-ATM parameters were subsequently optimized by joint grid searches. s_9 was fixed at 1.0 while $a_1 = 0.200$ – 0.500 was scanned in increments of 0.005 (61 values)

and $a_2 = 4.50\text{--}6.00$ in increments of 0.05 (31 values), for $61 \times 31 = 1,891$ combinations; this grid yielded the final values $a_1 = 0.215$ and $a_2 = 5.80$.

4.4.2 Effect of range-separation parameter ω

The range-separation parameter ω controls the partitioning between short-range (SR) and long-range (LR) Hartree-Fock exchange (HFX). A larger value of ω corresponds to a more rapid transition from SRHFX to LRHFX. Because we employ 100% LRHFX, increasing ω also implies a larger overall fraction of HFX at a given inter-electronic distance. Consequently, ω is a particularly important parameter in range-separated hybrid functionals.

Figure 9 summarizes the performance of COACH candidates across a range of ω values ($\omega = \{0.04, 0.08, 0.12, 0.16, 0.20, 0.22, 0.24, 0.26, 0.28, 0.30, 0.35, 0.40, 0.60\}$). For each ω , we recomputed the SRHFX and LRHFX contributions and adjusted the corresponding terms in the linear fitting problem, while keeping all other nonlinear parameters fixed. Performance was evaluated using the same NER-based metric as in the previous subsections.

	0.040	0.080	0.120	0.160	0.200	0.220	0.240	0.260	0.280	0.300	0.350	0.400	0.600
SRHFX	0.472	0.451	0.420	0.366	0.318	0.277	0.242	0.207	0.184	0.162	0.086	0.017	0.006
OF	0.902	0.881	0.852	1.101	1.090	1.026	0.954	1.000	1.078	1.031	1.022	1.219	2.068
BH	0.956	0.930	0.934	0.937	0.921	0.933	0.926	0.930	0.929	0.951	0.991	1.038	1.296
EF	1.135	0.987	0.981	0.994	1.075	1.075	1.140	1.243	1.085	1.195	1.285	1.374	1.672
FREQ	1.045	1.038	1.028	1.002	0.979	0.915	0.924	0.939	0.902	0.886	0.894	0.875	1.032
ISO	1.806	1.835	1.908	1.719	1.474	1.495	1.503	1.398	1.386	1.329	1.396	1.527	2.120
NC	1.984	1.746	1.644	1.511	1.371	1.279	1.257	1.201	1.148	1.135	1.107	1.106	1.452
TC	1.198	1.188	1.152	1.140	1.049	0.997	1.022	1.008	0.999	0.983	1.031	1.110	1.801
TM	1.134	1.164	1.160	1.198	1.163	1.185	1.195	1.232	1.223	1.216	1.266	1.286	1.580
Mean	1.275	1.231	1.199	1.159	1.091	1.069	1.065	1.055	1.039	1.038	1.063	1.112	1.476
Median	1.014	1.006	0.981	0.956	0.941	0.931	0.932	0.941	0.943	0.956	0.984	1.009	1.246

Fig. 9 Performance of the functional for different values of the range-separation parameter ω . “SRHFX” denotes the average fraction of short-range Hartree-Fock exchange for the five best functional candidates.

An interesting feature is the trend in the fraction of SRHFX as ω varies. When ω is very small, HFX is dominated by the short-range component, such that $c_{x,\text{sr}}^{\text{HF}} E_{x,\text{sr}}^{\text{HF}} + c_{x,\text{lr}}^{\text{HF}} E_{x,\text{lr}}^{\text{HF}} \approx c_{x,\text{sr}}^{\text{HF}} E_x^{\text{HF}}$. In this regime, the best-performing functionals favor larger values of $c_{x,\text{sr}}$, consistent with the high HFX fractions used in global hybrid functionals such as M06-2X and BMK. In contrast, when ω is large, HFX is dominated by the long-range component. Because 100% LRHFX is employed, a smaller $c_{x,\text{sr}}$ is then required to prevent the overall HFX contribution from becoming excessively large. When ω becomes very large, however, even values of $c_{x,\text{sr}}$ approaching zero are insufficient, and the functional effectively behaves as a 100% HFX hybrid (e.g., HFLYP), leading to very poor performance across nearly all systems.

At $\omega = 0.28$, the functional candidates achieve the best overall mean NER and perform well for most data categories, with the exception of OF and TM, for which smaller ω and larger $c_{x,\text{sr}}$ values are preferred. This behavior may reflect the greater importance of higher HFX fractions for systems containing heavier elements. The best overall median metric is observed at $\omega = 0.22$. Accordingly, we assessed the seven grid

points $\omega = \{0.22, 0.24, 0.25, 0.26, 0.27, 0.28, 0.29\}$ in the final refinement and selected $\omega = 0.27$ based on the overall data, noting that performance changes only weakly when ω varies between 0.24 and 0.29.

4.4.3 Effect of the same-spin correlation nonlinearity parameter $\gamma_{c,ss}$

The parameter $\gamma_{c,ss}$ appears in the finite-domain transformation of the reduced density gradient s^2 (Eqn. 10) used in the same-spin correlation fitting factor. In the original B97 formulation and in ω B97M-V, Becke’s transformation with $\gamma_{c,ss} = 0.2$ was adopted. As discussed above, recent work has shown that this historical value can lead to oscillatory functional derivatives and reduced numerical stability along potential-energy surfaces [12]. That study further demonstrated that reducing $\gamma_{c,ss}$ to 0.05 largely mitigates this problem, while $\gamma_{c,ss} = 0.005$ can make the oscillations nearly unobservable.

In addition, Ref. 12 proposed using an error-function-based finite-domain transformation,

$$u_{c,\sigma\sigma} = \text{erf}(\gamma_{c,ss}s_\sigma), \quad (53)$$

which can yield grid-insensitive derivative properties. However, this transformation introduces a non-zero derivative of the power-series expansion at the origin, requiring the first-order fitting coefficient to be forcibly set to zero in order to satisfy the uniform-electron-gas (UEG) constraint. Motivated by this consideration, we also tested an alternative form based on s_σ^2 ,

$$u_{c,\sigma\sigma} = \text{erf}(\gamma_{c,ss}s_\sigma^2), \quad (54)$$

which preserves the vanishing first derivative at zero.

Figure 10 summarizes the performance of COACH candidates obtained using different finite-domain transformation forms and values of $\gamma_{c,ss}$ in the final training cycle. The transformation screen used four grid points for each form. We used $\gamma_{c,ss} = \{0.005, 0.01, 0.02, 0.04\}$ for Becke’s original form, $\{0.03, 0.05, 0.10, 0.20\}$ for $\text{erf}(\gamma_{c,ss}s_\sigma)$, and $\{0.005, 0.01, 0.03, 0.05\}$ for $\text{erf}(\gamma_{c,ss}s_\sigma^2)$. For each setting, we recomputed the corresponding dimensionless variables, updated the matrix $\mathbf{A}$ in the linear fitting problem, and evaluated performance using the same NER-based metric employed throughout this section. NERs were averaged within each property category and over the full database.

Based on the overall mean and median NER metrics, the best performance is achieved using Becke’s original transformation with $\gamma_{c,ss} = 0.01$ –0.02. The best candidates using $\text{erf}(\gamma_{c,ss}s_\sigma)$ with $\gamma_{c,ss} = 0.2$ and $\text{erf}(\gamma_{c,ss}s_\sigma^2)$ with $\gamma_{c,ss} = 0.03$ yield slightly worse overall performance, with the mean NER increasing by less than 0.01. However, the degradation for transition-metal systems is more pronounced, with increases of up to 0.05. Accordingly, we retain Becke’s original functional form for the s^2 transformation in same-spin correlation and adopt the second-best but smaller value $\gamma_{c,ss} = 0.01$ in the final functional.

	$u_{c,ss} = \frac{\gamma_{c,ss}s_a^2}{1 + \gamma_{c,ss}s_a^2}$				$u_{c,ss} = \text{erf}(\gamma_{c,ss}s_a)$				$u_{c,ss} = \text{erf}(\gamma_{c,ss}s_a^2)$			
$\gamma_{c,ss}$	0.005	0.01	0.02	0.04	0.03	0.05	0.1	0.2	0.005	0.01	0.03	0.05
Mean	0.945	0.925	0.920	0.940	0.954	0.937	0.933	0.930	0.953	0.964	0.935	0.953
Median	0.859	0.847	0.845	0.852	0.854	0.852	0.857	0.857	0.848	0.873	0.850	0.854
H2O	0.658	0.638	0.589	0.582	0.691	0.682	0.575	0.609	0.666	0.755	0.627	0.698
OF	0.910	0.953	1.031	1.245	0.916	0.779	1.057	0.756	1.108	1.223	1.090	0.998
BH	0.872	0.851	0.868	0.857	0.852	0.860	0.861	0.843	0.848	0.859	0.860	0.855
EF	1.126	1.121	1.118	1.044	1.233	1.088	1.025	1.136	1.036	1.062	1.049	1.039
FREQ	0.921	0.924	0.918	0.954	0.969	0.962	0.952	0.934	0.934	0.940	0.937	0.953
ISO	1.045	1.034	0.985	1.019	1.070	1.034	1.042	1.061	1.131	1.136	0.994	1.085
NC	0.919	0.877	0.878	0.896	0.922	0.924	0.885	0.870	0.891	0.906	0.911	0.897
TC	0.887	0.880	0.886	0.921	0.884	0.873	0.896	0.875	0.906	0.915	0.902	0.914
TM	1.043	1.030	1.042	1.054	1.082	1.048	1.060	1.091	1.054	1.059	1.083	1.112

Fig. 10 Performance of the functional for different treatments of the same-spin correlation nonlinearity. Results compare Becke’s original transformation and two alternative finite-domain transformations with varying $\gamma_{c,ss}$ values.

5 Final functional form and all parameters

This section summarizes the final COACH functional form and all linear and nonlinear parameters used in the final model. As documented below, there are 73 independent linear parameters in total, as well as 3 reoptimized non-linear parameters and an optimized 3-body D4 correction. A total of 7 exact conditions are fully satisfied, and 5 more are partially, on the sampled mesh, or at the component level satisfied. In addition, adequate numerical stability was enforced.

The total exchange–correlation energy is written as

$$E_{xc} = E_x + E_c, \quad (55)$$

$$E_x = E_{x,sl}^{\text{SL}} + c_{x,sl}^{\text{HF}} E_{x,sl}^{\text{HF}} + c_{x,lr}^{\text{HF}} E_{x,lr}^{\text{HF}}, \quad (56)$$

$$E_c = E_{c,ss}^{\text{SL}} + E_{c,os}^{\text{SL}} + E_c^{\text{VV10}} + E_c^{\text{D4-ATM}}. \quad (57)$$

The range-separated Hartree–Fock exchange terms are defined in Eqs. 3 and 4. We fix $c_{x,lr}^{\text{HF}} = 1$, while the trained value of $c_{x,sl}^{\text{HF}}$ is 0.22878980716640696. The range-separation parameter is trained to $\omega = 0.27$.

The final dispersion treatment consists of a two-body nonlocal VV10 correlation term and a three-body D4-ATM correction. The VV10 parameters are

$$b = 5.5, \quad C = 0.01. \quad (58)$$

For the D4-ATM three-body contribution, we use the parameter set

$$s_6 = 0, \quad s_8 = 0, \quad s_9 = 1.0, \quad a_1 = 0.215, \quad a_2 = 5.8, \quad (59)$$

which was optimized on the L14 and vL11 benchmark sets [45] without significantly degrading performance on GSCDB137.

The semi-local exchange and correlation components share the general B97-type form

$$E_{x(c)}^{\text{SL}} = \int e_{x(c)}^{\text{base}} F_{\text{corr}} g_{x(c)} d\mathbf{r}. \quad (60)$$

Below, we summarize the base energy density $e_{x(c)}^{\text{base}}$, the multiplicative correction factor F_{corr} , and the fitting factor $g_{x(c)}$ for exchange, same-spin correlation, and opposite-spin correlation.

5.1 Semi-local (short-range) exchange

For exchange, we use UEG exchange energy density as the base,

$$e_{x,\sigma}^{\text{UEG}} = -\frac{3}{2} \left(\frac{3}{4\pi} \right)^{1/3} \rho_{\sigma}^{4/3}, \quad (61)$$

and apply the short-range correction factor

$$F_{x,\sigma}^{\text{sr}} = 1 - \frac{2}{3} a_{\sigma} \left(2\sqrt{\pi} \operatorname{erf} \left(\frac{1}{a_{\sigma}} \right) - 3a_{\sigma} + a_{\sigma}^3 + (2a_{\sigma} - a_{\sigma}^3) \exp \left(-\frac{1}{a_{\sigma}^2} \right) \right), \quad (62)$$

where $a_{\sigma} = \omega/k_{F_{\sigma}}$ and $k_{F_{\sigma}} = (6\pi^2 \rho_{\sigma})^{1/3}$.

We define the dimensionless variables $u_{x,\sigma}$ and $v_{\sigma} = 2\beta_{\sigma} - 1$ as

$$s_{\sigma} = \frac{|\nabla \rho_{\sigma}|}{\rho_{\sigma}^{4/3}}, \quad (63)$$

$$u_{x,\sigma} = \frac{\gamma_x s_{\sigma}^2}{1 + \gamma_x s_{\sigma}^2}, \quad (64)$$

$$\beta_{\sigma} = \frac{\tau_{\sigma} - \tau_{\sigma}^W}{\tau_{\sigma} + \tau_{\sigma}^{\text{UEG}}}, \quad (65)$$

$$\tau_{\sigma}^{\text{UEG}} = \frac{3}{5} (6\pi^2)^{2/3} \rho_{\sigma}^{5/3}, \quad (66)$$

$$\tau_{\sigma}^W = \frac{1}{4} \frac{|\nabla \rho_{\sigma}|^2}{\rho_{\sigma}}, \quad (67)$$

with $\gamma_x = 0.004$.

The exchange fitting factor $g_{x,\sigma}$ is expanded using Legendre polynomials in v_{σ} and monomials in $u_{x,\sigma}$,

$$g_{x,\sigma} = g_{x,\sigma}(\nabla \rho_{\sigma}, \tau_{\sigma}) = \sum_{i=0}^{12} \sum_{j=0}^8 c_{x,ij} f_i(v_{\sigma}) f_j(u_{x,\sigma}), \quad (68)$$

where $f_i(v) \equiv P_i(v)$ are the (standard) Legendre polynomials on $[-1, 1]$ with $P_0(v) = 1$, $P_1(v) = v$, and $f_j(u) \equiv u^j$ are monomials on $[0, 1]$. The nonzero coefficients are listed in Table 3.

Table 3 Nonzero exchange expansion coefficients.

$c_{i,j}$	$c_{i,j}$
$c_{0,0} = 0.7080650005052257$	$c_{5,2} = -3.463397249605392$
$c_{0,2} = 5.973423554868972$	$c_{5,3} = 3.75514691066743$
$c_{0,3} = -17.33477136180401$	$c_{6,0} = 0.2402820743898442$
$c_{0,5} = 25.0$	$c_{6,1} = -0.8660852096125707$
$c_{0,7} = -13.303231846931668$	$c_{7,0} = 0.10480172805257999$
$c_{1,0} = -0.20577489944575586$	$c_{7,1} = -0.5657722952687481$
$c_{2,0} = -0.10779574940300828$	$c_{8,1} = -0.43586821950068677$
$c_{2,2} = 5.491317004106042$	$c_{8,4} = 0.9891478454228174$
$c_{2,3} = -6.658515516611359$	$c_{9,0} = -0.09350557083826855$
$c_{3,1} = 1.839797240566924$	$c_{9,1} = 0.2557276229171211$
$c_{3,2} = -7.509521330876057$	$c_{9,2} = -0.7171538898115716$
$c_{3,3} = 18.654513589197446$	$c_{10,0} = -0.034948386270515386$
$c_{3,4} = -12.139029019506687$	$c_{11,0} = -0.03464312342837082$
$c_{4,0} = 0.20195969717314624$	$c_{11,1} = 0.20050993281346377$
$c_{5,0} = 0.2600717991008879$	$c_{11,2} = -0.3773473250434576$

5.2 Same-spin correlation

For correlation, we adopt SCAN correlation ($\alpha = 1$) as the base functional for both same-spin and opposite-spin components. In the $\alpha = 1$ limit, SCAN correlation reduces to a revised PBE-like form,

$$\varepsilon_c^1 = \varepsilon_c^{\text{LSDA1}} + H_1, \quad (69)$$

$$H_1 = \gamma \phi^3 \ln[1 + w_1 (1 - g(At^2))], \quad (70)$$

$$w_1 = \exp\left[-\frac{\varepsilon_c^{\text{LSDA1}}}{\gamma \phi^3}\right] - 1, \quad (71)$$

$$g(At^2) = \frac{1}{(1 + 4At^2)^{1/4}}, \quad (72)$$

$$A = \frac{\beta(r_s)}{\gamma w_1}, \quad (73)$$

$$t = \left(\frac{3\pi^2}{16}\right)^{1/3} \frac{s}{\phi r_s^{1/2}}, \quad (74)$$

$$s = \frac{|\nabla\rho|}{(2(3\pi^2)^{1/3}\rho^{4/3})}. \quad (75)$$

Here $\varepsilon_c^{\text{LSDA1}}$ denotes the LSDA correlation energy density (we use PW92), $\gamma = 0.031091$,

$$\beta(r_s) = 0.066725 \frac{1 + 0.1r_s}{1 + 0.1778r_s}, \quad \phi = \frac{(1 + \zeta)^{2/3} + (1 - \zeta)^{2/3}}{2}, \quad (76)$$

with $r_s = \left(\frac{3}{4\pi\rho}\right)^{1/3}$ and $\zeta = (\rho_\alpha - \rho_\beta)/\rho$.

The same-spin and opposite-spin base correlation energy densities are defined as

$$e_{c,\sigma\sigma}^{\text{base}} = e_c^{\text{base}}(\rho_\sigma, 0), \quad (77)$$

$$e_{c,os}^{\text{base}} = e_c^{\text{base}}(\rho_\alpha, \rho_\beta) - e_c^{\text{base}}(\rho_\alpha, 0) - e_c^{\text{base}}(0, \rho_\beta). \quad (78)$$

To enforce the self-correlation correction (SCC), we use the multiplicative correction factor for same-spin correlation.

$$F_{c,ss} = 2\beta_\sigma. \quad (79)$$

For same-spin correlation, we use the dimensionless variables $v_\sigma = 2\beta_\sigma - 1$ (the same variable as the exchange) and $u_{c,\sigma\sigma}$, defined as

$$u_{c,\sigma\sigma} = \frac{\gamma_{c,ss}s_\sigma^2}{1 + \gamma_{c,ss}s_\sigma^2}. \quad (80)$$

with $\gamma_{c,ss} = 0.01$.

The fitting factor $g_{c,\sigma\sigma}$ is expanded using Legendre polynomials in v_σ and monomials in $u_{c,\sigma\sigma}$, with nonzero coefficients listed in Table 4.

Table 4 Nonzero same-spin expansion coefficients.

$c_{i,j}$	$c_{i,j}$
$c_{0,0} = -0.3039405655250467$	$c_{3,1} = -2.226676731002602$
$c_{0,1} = -1.2900096158738523$	$c_{3,2} = -2.7311170048590196$
$c_{0,4} = 17.715459474491766$	$c_{5,0} = -1.4633617968363573$
$c_{0,7} = -13.929080028872658$	$c_{7,0} = -0.8348281349925031$
$c_{1,0} = -1.8711635396581212$	$c_{8,0} = 0.4818154118907339$
$c_{1,1} = -10.122833804966692$	$c_{8,1} = -3.3444231176840256$
$c_{1,2} = 22.05988705553151$	$c_{8,2} = 6.986341349369924$
$c_{2,0} = -1.8278556540465263$	$c_{8,3} = 14.980984577715752$
$c_{2,1} = 6.994998018578696$	$c_{8,4} = -25.0$
$c_{2,4} = -8.349141007699943$	$c_{11,3} = 1.5676639733349753$

5.3 Opposite-spin correlation

For opposite-spin correlation, we use the same base correlation functional defined above. The opposite-spin base contribution is given in Eq. 78. There is no the multiplicative correction factor for the opposite-spin correlation energy.

Then we use dimensionless variables $u_{c,\alpha\beta}$ and $w_{c,\alpha\beta}$, defined as

$$s_{\alpha\beta}^2 = \frac{1}{2}(s_\alpha^2 + s_\beta^2), \quad (81)$$

$$u_{c,\alpha\beta} = \frac{\gamma_{c,os}s_{\alpha\beta}^2}{1 + \gamma_{c,os}s_{\alpha\beta}^2}, \quad (82)$$

$$w_{c,\alpha\beta} = \frac{t_{\alpha\beta} - 1}{t_{\alpha\beta} + 1}, \quad (83)$$

$$t_{\alpha\beta} = \frac{1}{2}(t_{\alpha} + t_{\beta}), \quad (84)$$

$$t_{\sigma} = \frac{\tau_{\sigma}^{\text{UEG}}}{\tau_{\sigma}}. \quad (85)$$

with $\gamma_{c,os} = 0.006$, identical to the value used in $\omega\text{B97M-V}$.

The opposite-spin fitting factor is expanded using Legendre polynomials in both $w_{c,\alpha\beta}$ and $u_{c,\alpha\beta}$. The nonzero expansion coefficients are listed in Table 5.

Table 5 Nonzero opposite-spin expansion coefficients.

$c_{i,j}$	$c_{i,j}$
$c_{0,0} = 2.0330833570991405$	$c_{5,6} = -22.533887529213906$
$c_{0,5} = -3.071472897617402$	$c_{5,7} = 9.590467389102345$
$c_{0,7} = -1.9671740896453935$	$c_{6,4} = 4.4188486804036184$
$c_{1,0} = -3.124170248419258$	$c_{8,4} = 4.293474773028771$
$c_{1,1} = 7.594716450411337$	$c_{9,2} = -1.5286479621022213$
$c_{2,4} = 4.618674090063557$	$c_{9,7} = 2.3555422987546892$
$c_{3,7} = 2.572431599509026$	$c_{10,0} = 0.7896380110221484$
$c_{4,4} = -1.120055362237976$	$c_{10,3} = -1.9225815901087469$
$c_{4,7} = -2.5762282779751264$	$c_{11,1} = 2.229310613916473$
$c_{5,1} = -1.4299548850828665$	$c_{11,6} = -1.5811003317333285$
$c_{5,4} = -20.76844377770899$	$c_{11,7} = 1.1456216634190832$
$c_{5,5} = 25.0$	

6 Extended post-development evaluation of COACH

6.1 Behavior of the final COACH inhomogeneity correction factors

After the functional form and coefficients were fixed, we examined the final COACH inhomogeneity correction factors (ICFs) over the bounded domains $0 \leq u \leq 1$ and $-1 \leq v, w \leq 1$. Figure 11 shows the exchange factor $G_x = F_x^{\text{NUSC}} g_x$ and the same-spin and opposite-spin correlation factors, $g_{c,ss}$ and $g_{c,os}$.

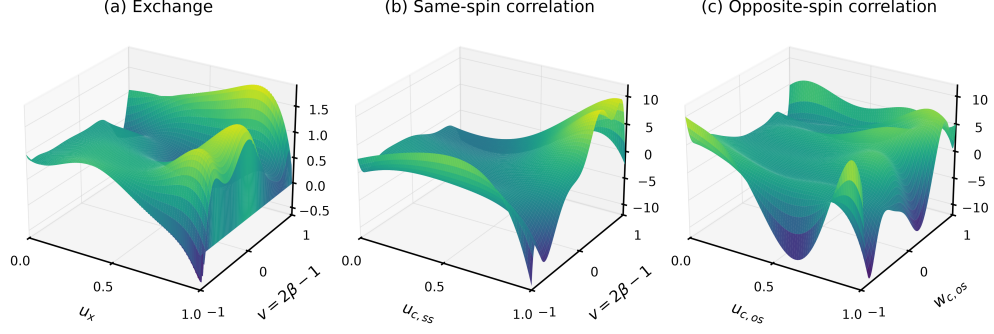


Fig. 11 Final COACH ICF surfaces over the complete bounded variable domains. Panel (a) shows $G_x = F_x^{\text{NUSC}} g_x$, and panels (b) and (c) show $g_{c,ss}$ and $g_{c,os}$, respectively.

Over the complete domains, the correlation factors span $-11.262421174 \leq g_{c,ss} \leq 10.815183096$ and $-11.119961775 \leq g_{c,os} \leq 10.021487957$. To assess the exchange factor in physically sampled regions, we performed self-consistent COACH/def2-SVP calculations for the 18 neutral atoms H–Ar using an unpruned PySCF (99, 590) grid. Each spin-resolved point was assigned the short-range exchange weight

$$W_i = |w_i e_x^{\text{UEG}}(\rho_{\sigma i}) F_{x,\text{sr}}(\rho_{\sigma i})|. \quad (86)$$

For each atom separately, points were ordered by W_i , and the smallest set reaching the specified fraction of that atom’s exchange weight was retained. Table 6 compares the resulting ranges with the optimization target $0 \leq G_x \leq 2.2146 g_x^{\text{UEG}} = 1.7079$ and with the complete variable domain. It shows that constraint satisfaction on the optimization mesh does not extend to the complete domain. However, the 99.99% AE18 union lies within the target interval, indicating that X01 and XC14 are satisfied in the physically relevant region. We similarly assessed the semi-local X06 upper bound, which is satisfied over the complete domain.

Table 6 Range of the final COACH exchange factor G_x under the target constraint, on the optimization mesh, for the AE18 atomic densities, and over the complete bounded-variable domain. The AE18 rows are unions of point sets selected independently for each atom to reach the indicated fraction of its exchange weight.

Sampling region	Minimum G_x	Maximum G_x
Target interval (X01, XC14)	0.0000	1.7079 ($= 2.2146 g_x^{\text{UEG}}$)
Optimization constraint mesh	0.0000	1.7079
AE18, per-atom 99.99% weight coverage	0.2396	1.3773
AE18, per-atom 99.999% weight coverage	−0.2175	1.6578
Complete bounded-variable domain	−0.5525	1.7922

6.2 One-electron self-interaction diagnostics for H and H₂⁺

We evaluated the H atom and H₂⁺ using COACH, ω B97M-V, and HF with the def2-QZVPPD basis. For the H atom, we report the individual energy contributions and $E + 0.5$, measured relative to the exact nonrelativistic H energy.

Table 7 One-electron H-atom diagnostics for COACH, HF, and ω B97M-V in the def2-QZVPPD basis. All quantities are in kcal mol⁻¹.

Method	$J + E_x$	E_c^{SL}	E_c^{VV10}	$E + 0.5$
COACH	-1.4924	+0.0000	+3.2249	+2.0551
ω B97M-V	-2.7793	+2.0992	+2.8412	+3.3945
HF	+0.0000	+0.0000	+0.0000	+0.0105

As expected for a one-electron HF calculation, $J + E_x$ vanishes to numerical precision, and $E + 0.5$ is 0.0105 kcal mol⁻¹. For COACH, $J + E_x = -1.492$ kcal mol⁻¹, and E_c^{SL} is exactly zero because the functional satisfies the one-electron zero-correlation constraint C11. The separately evaluated VV10 contribution is 3.225 kcal mol⁻¹, giving $J + E_{xc} = 1.733$ kcal mol⁻¹. The resulting net residual is 0.429 kcal mol⁻¹ smaller than that of ω B97M-V, demonstrating the improvement provided by the semilocal exchange and correlation treatments.

The H₂⁺ scan comprises 11 fixed H–H distances spanning 1.0573–10.0 Å. At each distance, the energy is referenced to that of the H atom computed separately with the same method and def2-QZVPPD basis: $\Delta E(R) = E(\text{H}_2^+, R) - E(\text{H})$. The negative values at short range include the physical H₂⁺ bonding contribution, whereas the long-range limit should approach zero for an exact one-electron treatment.

At 10 Å, ΔE is -18.688 kcal mol⁻¹ for COACH and -24.812 kcal mol⁻¹ for ω B97M-V, whereas HF gives -0.011 kcal mol⁻¹. The HF result confirms that finite-basis and finite-distance effects are negligible on this scale. Panel 12b shows that the COACH semilocal correlation is exactly zero throughout the one-electron scan, as required by C11. From 1.0573 to 10 Å, $J + E_x$ becomes increasingly negative, decreasing from -4.791 to -21.033 kcal mol⁻¹, whereas E_c^{VV10} changes only from +3.265 to +3.310 kcal mol⁻¹. The remaining one-electron and nuclear-repulsion contributions sum to only +0.767 kcal mol⁻¹, showing that the long-range plateau is dominated by the negative $J + E_x$ term and is only partially compensated by VV10. COACH reduces the magnitude of the ω B97M-V long-range error by 6.124 kcal mol⁻¹, or approximately 25%, although it retains a finite negative offset. Taken together, the H and H₂⁺ results show a modest reduction in one-electron self-interaction and charge-delocalization errors relative to ω B97M-V.

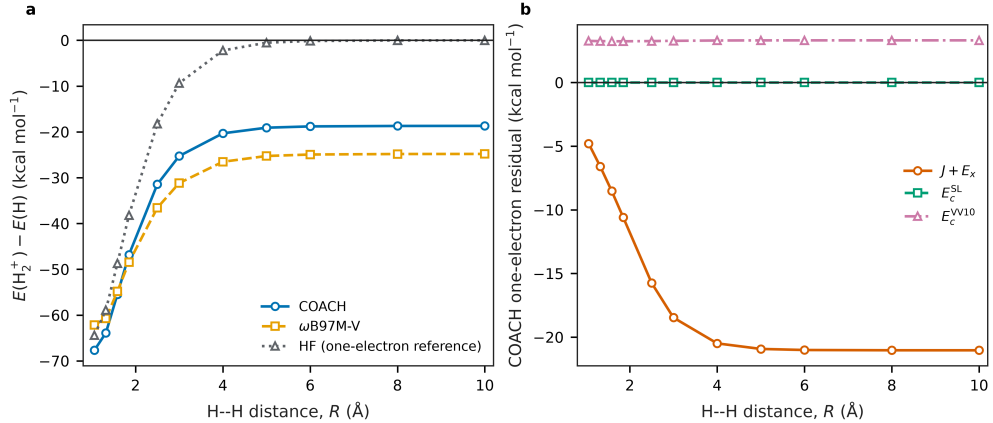


Fig. 12 One-electron H_2^+ dissociation diagnostics in the def2-QZVPPD basis. **(a)** $E(\text{H}_2^+) - E(\text{H})$ for COACH, HF, and $\omega\text{B97M-V}$; the horizontal zero line is the exact dissociation limit. **(b)** Decomposition of the COACH one-electron residual into $J + E_x$, E_c^{SL} , and E_c^{VV10} . Lines connect the 11 explicitly calculated distances without interpolation or smoothing.

6.3 Atomic HOMO energies, ΔSCF ionization energies, and Koopmans deviations

We further evaluated the first 15 atomic ionization reactions in G21IP, comprising H-F and Na-Cl, using the same aug-pc-4 neutral and cation inputs for unrestricted COACH, $\omega\text{B97M-V}$, and HF calculations. For H^+ , the zero-electron energy is zero; for every other atom, the ionization energy was obtained from independently converged neutral and cation calculations. We define

$$I_{\Delta\text{SCF}} = E(N-1) - E(N), \quad \Delta_K = -\varepsilon_{\text{HOMO}} - I_{\Delta\text{SCF}}. \quad (87)$$

Table 8 Aggregate errors for the 15 atomic G21IP reactions. All statistics are in kcal mol^{-1} .

Method	IP MSE	IP MAE	IP RMSE	IP maxAE	Δ_K MSE	Δ_K MAE
COACH	+0.378	2.518	2.722	4.520	-22.800	22.800
HF	-17.203	17.203	21.212	41.246	+20.267	20.267
$\omega\text{B97M-V}$	-1.461	2.659	3.632	6.720	-25.500	25.500

The ΔSCF IP MAE is $2.518 \text{ kcal mol}^{-1}$ for COACH and $2.659 \text{ kcal mol}^{-1}$ for $\omega\text{B97M-V}$, whereas the corresponding $|\Delta_K|$ MAEs are 22.800 and $25.500 \text{ kcal mol}^{-1}$, respectively. For this 15-reaction atomic subset, these results show that COACH provides more accurate IP predictions than $\omega\text{B97M-V}$ and yields HOMO energies that are more consistent with the corresponding ΔSCF ionization energies. By contrast, HF

gives a much larger ΔSCF IP MAE of 17.203 kcal mol⁻¹ but a $|\Delta_K|$ MAE of 20.267 kcal mol⁻¹, consistent with the distinct frozen-orbital content of the HF Koopmans relation.

6.4 Basis-set convergence relative to the CBS proxy

Figure 4 of the main text reports errors relative to theoretical best estimates, whereas here each available finite-basis result is compared with the current recommended-basis COACH value (used as the CBS proxy) for the same reaction or property. Following the GSCDB139 recommended-basis protocol, the reference basis is aug-pc-3 for HR46, def2-QZVPP for MOR13, TMB11, and TMD10, and def2-QZVPPD for TAE_W4-17nonMR, MB16-43, and S66. The top-panel NERs in Figure 13 are calculated using the same data-set-specific metrics and normalization as in Figure 4. The bottom panel instead reports the RMSEs of the MOR13, TMB11, and TMD10 basis deviations in kcal mol⁻¹, thereby avoiding the fixed residual used in the NER calculations.

For all 30 displayed small-to-large basis sequences, the errors relative to the current recommended-basis results are nonincreasing, using NER in the top panel and RMSE in the bottom panel. The same trend holds outside the def2 family: for every displayed data set, (aug-)pc-3 is closer to the current reference than (aug-)pc-2, and (aug-)cc-pVQZ is closer than (aug-)cc-pVTZ. By contrast, the errors relative to theoretical best estimates in Figure 4 contain both functional and basis-set contributions, allowing a smaller basis to occasionally yield a lower benchmark error through cancellation between them.

BASIS	TAE_W4-17nonMR	MB16-43	S66	HR46
def2-QZVPPD	0.00	0.00	0.00	0.23
def2-QZVPP	0.04	0.07	0.42	4.07
def2-TZVPD	0.23	0.25	0.59	0.79
def2-TZVP	0.33	0.27	2.35	9.10
def2-SVPD	2.75	1.36	8.62	2.80
def2-SVP	2.83	1.19	11.17	19.87
pc-3	0.10	0.05	0.16	2.15
aug-pc-3	0.11	0.04	0.12	0.00
aug-pc-2	0.34	0.16	1.32	0.14
pc-2	0.50	0.25	1.60	8.48
aug-pc-1	2.54	0.70	9.58	0.85
pc-1	3.80	0.81	9.50	18.93
aug-cc-pVQZ	0.33	0.13	0.49	0.10
cc-pVQZ	0.39	0.19	1.99	5.62
aug-cc-pVTZ	0.66	0.22	1.05	0.51
cc-pVTZ	0.80	0.30	4.49	10.53
aug-cc-pVDZ	4.08	1.07	4.46	1.37
cc-pVDZ	4.17	0.97	10.50	19.49

BASIS	MOR13	TMB11	TMD10
def2-QZVPPD	0.05	0.09	0.02
def2-QZVPP	0.00	0.00	0.00
def2-TZVPD	0.70	0.20	1.20
def2-TZVP	0.76	0.55	2.24
def2-SVPD	5.36	2.60	5.70
def2-SVP	4.93	3.68	6.98

Fig. 13 Basis-set convergence of COACH relative to the current recommended-basis results. **Top:** NERs of the basis deviations for TAE_W4-17nonMR, MB16-43, S66, and HR46 across the same def2, pc, and cc basis-set ranges shown in Figure 4. **Bottom:** RMSEs of the basis deviations for MOR13, TMB11, and TMD10 in kcal mol⁻¹ across the same def2 basis-set range.

6.5 Evaluation protocol on additional sets

The external evaluation was conducted only after the final COACH form and parameters had been fixed, and none of the results reported in this section was used for refitting. For each benchmark, we report the MAE, MSE, and RMSE over the complete set without excluding outliers. Consistent with the setup of the source study, H2X100 binding energies and stoichiometric coefficients are normalized by the number of pairwise interactions; for TAE-PTComp, both quantities are divided by the molecular atom count. The reference values were taken directly from the original benchmark studies [46–54]. When a source study reports DFT benchmarks, we select its best-performing reported functional as the comparator and follow the corresponding basis-set protocol; otherwise, we use ω B97M-V as the comparator and employ a setup consistent with GSCDB137. Specifically, we use def2-TZVP for GDB9-W1-F12; def2-TZVPPD with counterpoise correction for vL27; def2-TZVPPD for W1-SN2-BH; def2-QZVPP for 3dCIPEA, TRIP50, ISOC7, and TAE-PTComp; def2-QZVPPD with counterpoise correction for H2X100; and def2-QZVPPD for atmospheric218. Because 11 vL27 complexes also occur in the BigNC/vL11 design set, the full vL27 result is not wholly independent. However, the vL27 reference values for these systems differ from those used during fitting. The original vL11 references used DLPNO-CCSD(T1)/CBS with VeryTightPNO thresholds, whereas vL27 recomputed these systems using the more accurate 7/8 complete-PNO-space (CPS) extrapolation [54]. In the vL27 study, this refinement reduced the MAE against canonical CCSD(T)/def2-SVP for L14 from 0.39 to 0.11 kcal mol⁻¹. On the nonoverlapping vL16 subset, COACH gives MAE/MSE/RMSE values of 1.102/0.126/1.299 kcal mol⁻¹, compared with 1.387/0.299/1.560 kcal mol⁻¹ for the local ω B97M-V calculation augmented by the D4-ATM three-body contribution.

6.6 Element-resolved analysis of TAE-PTComp

TAE-PTComp contains 2,097 atom-scaled total atomization energies divided among 67 element-specific subsets spanning the periodic table [50]. For element X , we follow the source study and define

$$\text{sMAE}_X = \frac{1}{N_X} \sum_{i \in X} \left| \frac{\text{TAE}_i^{\text{calc}} - \text{TAE}_i^{\text{ref}}}{n_i} \right|, \quad (88)$$

where N_X is the number of molecules containing X , and n_i is the number of atoms in molecule i . This decomposition tests whether a low aggregate error conceals localized errors for particular elements.

COACH gives a mean sMAE of 0.74 kcal mol⁻¹ over the 67 element-specific subsets, compared with 0.85 kcal mol⁻¹ for the published ω B97M-V values. The COACH sMAE is lower for 45 of the 67 subsets. Eleven elements have a COACH sMAE greater than 1.0 kcal mol⁻¹, compared with 16 for the published ω B97M-V results. The largest reductions occur for V, W, Ta, Hf, Sb, and Ti, whereas the largest increases occur for Ba, Zn, and Li. The maximum COACH sMAE is 1.51 kcal mol⁻¹ for W, below the published ω B97M-V maximum of 2.1 kcal mol⁻¹ for the same element. Overall,

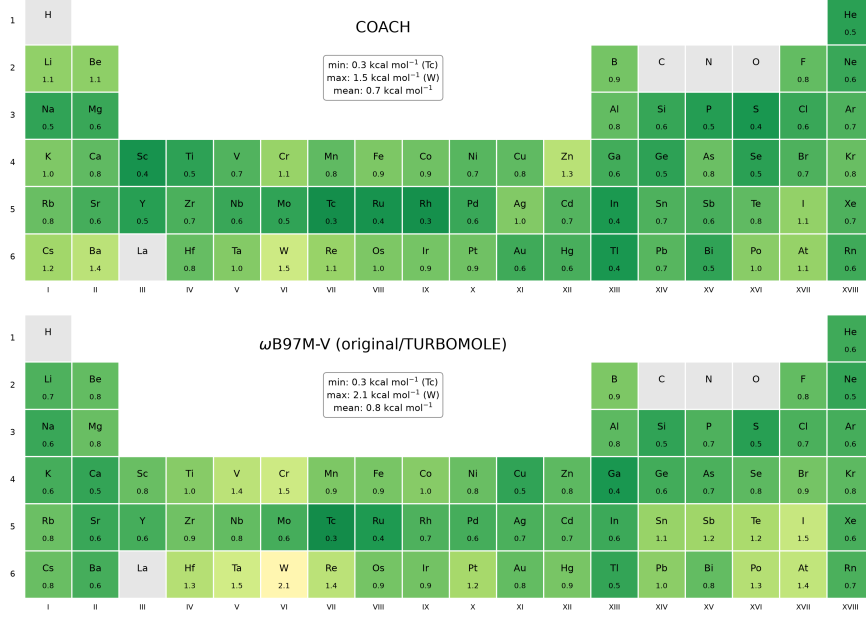


Fig. 14 Element-resolved scaled mean absolute errors (sMAEs) on TAE-PTComp. The panels show COACH computed with Q-Chem and the published ω B97M-V results computed with TURBOMOLE 8.0. Both calculations use def2-QZVPP with the appropriate def2 effective core potentials and the same 0–4 kcal mol^{−1} color scale. Blank cells denote elements outside the 67 subsets.

COACH improves the sMAE across most elements, with especially pronounced gains for the largest ω B97M-V outliers.

7 Further comments on nonlocal functional, the possible next generation of DFT

Our main-text conclusion notes that further substantial and systematic gains in both accuracy and transferability are likely to require going beyond semilocal density ingredients combined with simple global or range-separated exact-exchange mixing. Here we provide additional reasoning on this perspective.

The exact nonlocal exchange-correlation contribution from electron-electron interaction can be expressed from the pair density $n_2(\mathbf{r}_1, \mathbf{r}_2)$ as

$$n_{xc}^{nl} = \frac{n_2(\mathbf{r}_1, \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|} - \frac{\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (89)$$

$$n_2(\mathbf{r}_1, \mathbf{r}_2) = N(N-1) \int d\mathbf{x}_3 \cdots d\mathbf{x}_N |\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N)|^2. \quad (90)$$

One may then ask why the variational principle is not applied directly to the pair density to obtain the exact energy. The main difficulty is the N -representability problem: a

given pair density does not uniquely correspond to a physically admissible N -body wavefunction. However, by the Hohenberg–Kohn theorem, ground-state observables are functionals of the density $\rho(\mathbf{r})$, so $n_{xc}^{nl}(\mathbf{r}_1, \mathbf{r}_2) = n_{xc}^{nl}[\rho(\mathbf{r})]$.

This relation motivates approximating the exact exchange-correlation (or equivalently pair density) by one-electron density. In this perspective, one possible route is a generalized VV10-type representation at two spatial points:

$$n_{xc}^{nl}[\rho(\mathbf{r})](\mathbf{r}_1, \mathbf{r}_2) \approx n_{xc}^{nl}(\rho(\mathbf{r}_1), \nabla\rho(\mathbf{r}_1), \tau(\mathbf{r}_1), \rho(\mathbf{r}_2), \nabla\rho(\mathbf{r}_2), \tau(\mathbf{r}_2)), \quad (91)$$

This representation incorporates richer physical information and leads to a 10-dimensional optimization problem with substantially greater flexibility than the traditional two-dimensional DFT framework.

More generally, local-hybrid ideas [55] suggest an orbital-informed form,

$$n_{xc}^{nl}[\rho(\mathbf{r})](\mathbf{r}_1, \mathbf{r}_2) \approx n_{xc}^{nl}[\{\psi(\mathbf{r})\}](\mathbf{r}_1, \mathbf{r}_2) \approx n_{xc}^{nl}(\rho(\mathbf{r}_1), \{\psi(\mathbf{r}_1)\}, \rho(\mathbf{r}_2), \{\psi(\mathbf{r}_2)\}), \quad (92)$$

where $\{\psi(\mathbf{r})\}$ Kohn-Sham molecular orbitals.

Recent machine-learning approaches, such as the Skala functional [56], may also be viewed as belonging to this direction. Their framework uses neural-network architectures with nonlocal communication on coarse grids, implemented in practice at atomic centers, to incorporate additional nonlocal information.

We view these approaches as promising steps toward the next stage of DFT development, complementary to improvements in traditional double hybrids.

References

- [1] Becke, A. D. Density-functional thermochemistry. v. systematic optimization of exchange-correlation functionals. *J. Chem. Phys.* **107**, 8554–8560 (1997).
- [2] Becke, A. D. A new inhomogeneity parameter in density-functional theory. *J. Chem. Phys.* **109**, 2092–2098 (1998).
- [3] Mardirossian, N. & Head-Gordon, M. ω b97m-v: A combinatorially optimized, range-separated hybrid, meta-gga density functional with vv10 nonlocal correlation. *J. Chem. Phys.* **144**, 214110 (2016). URL <https://doi.org/10.1063/1.4952647>.
- [4] Iikura, H., Tsuneda, T., Yanai, T. & Hirao, K. A long-range correction scheme for generalized-gradient-approximation exchange functionals. *J. Chem. Phys.* **115**, 3540–3544 (2001).
- [5] Caldeweyher, E. *et al.* A generally applicable atomic-charge dependent london dispersion correction. *J. Chem. Phys.* **150** (2019).
- [6] Vydrov, O. A. & Van Voorhis, T. Nonlocal van der waals density functional: The simpler the better. *J. Chem. Phys.* **133**, 244103 (2010). URL <https://doi.org/10.1063/1.3521275>.

- [7] Gill, P. M., Adamson, R. D. & Pople, J. A. Coulomb-attenuated exchange energy density functionals. *Mol. Phys.* **88**, 1005–1009 (1996).
- [8] Leininger, T., Stoll, H., Werner, H.-J. & Savin, A. Combining long-range configuration interaction with short-range density functionals. *Chem. Phys. Lett.* **275**, 151–160 (1997).
- [9] Perdew, J. P. & Wang, Y. Accurate and simple analytic representation of the electron-gas correlation energy. *Phys. Rev. B* **45**, 13244–13249 (1992). URL <https://doi.org/10.1103/physrevb.45.13244>.
- [10] Sun, J., Ruzsinszky, A. & Perdew, J. P. Strongly constrained and appropriately normed semilocal density functional. *Phys. Rev. Lett.* **115**, 036402 (2015). URL <https://doi.org/10.1103/physrevlett.115.036402>.
- [11] Becke, A. D. Density functional calculations of molecular bond energies. *J. Chem. Phys.* **84**, 4524–4529 (1986).
- [12] Matito, E. Modern density functional approximations are ill-designed to compute vibrational properties. Invited lecture at WATOC 2025 (22nd World Association of Theoretical and Computational Chemists International Congress), Oslo, Norway (2025). June 22–27, 2025.
- [13] Furness, J. W. & Sun, J. Enhancing the efficiency of density functionals with an improved iso-orbital indicator. *Phys. Rev. B* **99**, 041119 (2019).
- [14] Mardirossian, N. & Head-Gordon, M. Mapping the genome of meta-generalized gradient approximation density functionals: The search for b97m-v. *J. Chem. Phys.* **142**, 074111 (2015). URL <https://doi.org/10.1063/1.4907719>.
- [15] Tao, J., Perdew, J. P., Staroverov, V. N. & Scuseria, G. E. Climbing the density functional ladder: Nonempirical meta-generalized gradient approximation designed for molecules and solids. *Phys. Rev. Lett.* **91**, 146401 (2003). URL <https://doi.org/10.1103/physrevlett.91.146401>.
- [16] Sparrow, Z. M., Ernst, B. G., Quady, T. K. & DiStasio Jr, R. A. Uniting nonempirical and empirical density functional approximation strategies using constraint-based regularization. *J. Phys. Chem. Lett.* **13**, 6896–6904 (2022).
- [17] Levy, M. & Perdew, J. P. Hellmann-feynman, virial, and scaling requisites for the exact universal density functionals. shape of the correlation potential and diamagnetic susceptibility for atoms. *Phys. Rev. A* **32**, 2010 (1985).
- [18] Svendsen, P.-S. & von Barth, U. Gradient expansion of the exchange energy from second-order density response theory. *Phys. Rev. B* **54**, 17402 (1996).

- [19] Antoniewicz, P. & Kleinman, L. Kohn-sham exchange potential exact to first order in $\rho(k \rightarrow 0)/\rho(0)$. *Phys. Rev. B* **31**, 6779 (1985).
- [20] Langreth, D. C. & Mehl, M. Beyond the local-density approximation in calculations of ground-state electronic properties. *Phys. Rev. B* **28**, 1809 (1983).
- [21] Becke, A. D. Density-functional exchange-energy approximation with correct asymptotic behavior. *Phys. Rev. A* **38**, 3098 (1988).
- [22] Levy, M. Density-functional exchange correlation through coordinate scaling in adiabatic connection and correlation hole. *Phys. Rev. A* **43**, 4637 (1991).
- [23] Görling, A. & Levy, M. Requirements for correlation energy density functionals from coordinate transformations. *Phys. Rev. A* **45**, 1509 (1992).
- [24] Lieb, E. H. & Oxford, S. Improved lower bound on the indirect coulomb energy. *Int. J. Quantum Chem.* **19**, 427–439 (1981).
- [25] Gadre, S. R., Bartolotti, L. J. & Handy, N. C. Bounds for coulomb energies. *J. Chem. Phys.* **72**, 1034–1038 (1980).
- [26] Perdew, J. P., Ruzsinszky, A., Sun, J. & Burke, K. Gedanken densities and exact constraints in density functional theory. *J. Chem. Phys.* **140**, 18A533 (2014).
- [27] Wang, Y. & Perdew, J. P. Spin scaling of the electron-gas correlation energy in the high-density limit. *Phys. Rev. B* **43**, 8911 (1991).
- [28] Kaplan, A. D., Levy, M. & Perdew, J. P. The predictive power of exact constraints and appropriate norms in density functional theory. *Annu. Rev. Phys. Chem.* **74**, 193–218 (2023). URL <https://doi.org/10.1146/annurev-physchem-062422-013259>.
- [29] Levy, M. & Perdew, J. P. Tight bound and convexity constraint on the exchange-correlation-energy functional in the low-density limit, and other formal tests of generalized-gradient approximations. *Phys. Rev. B* **48**, 11638 (1993).
- [30] Gräfenstein, J., Izotov, D. & Cremer, D. Avoiding singularity problems associated with meta-gga (generalized gradient approximation) exchange and correlation functionals containing the kinetic energy density. *J. Chem. Phys.* **127** (2007).
- [31] Pollack, L. & Perdew, J. Evaluating density functional performance for the quasi-two-dimensional electron gas. *J. Phys.: Condens. Matter* **12**, 1239 (2000).
- [32] Pople, J. A., Binkley, J. S. & Seeger, R. Theoretical models incorporating electron correlation. *Int. J. Quantum Chem.* **10**, 1–19 (1976).
- [33] Chan, G. K.-L. & Handy, N. C. Optimized lieb-oxford bound for the exchange-correlation energy. *Phys. Rev. A* **59**, 3075 (1999).

- [34] Odashima, M. M., Capelle, K. & Trickey, S. Tightened lieb-oxford bound for systems of fixed particle number. *J. Chem. Theory Comput.* **5**, 798–807 (2009).
- [35] Manna, D. & Martin, J. M. L. What are the ground state structures of c20 and c24? an explicitly correlated ab initio approach. *J. Phys. Chem. A* **120**, 153–160 (2016). URL <http://dx.doi.org/10.1021/acs.jpca.5b10266>.
- [36] Sitkiewicz, S. P., Zaleśny, R., Ramos-Cordoba, E., Luis, J. M. & Matito, E. How reliable are modern density functional approximations to simulate vibrational spectroscopies? *J. Phys. Chem. Lett.* **13**, 5963–5968 (2022).
- [37] Natarajan, B. K. Sparse approximate solutions to linear systems. *SIAM J. Comput.* **24**, 227–234 (1995).
- [38] Bertsimas, D., King, A. & Mazumder, R. Best subset selection via a modern optimization lens. *Ann. Stat.* 813–852 (2016).
- [39] Gurobi Optimization, LLC. Gurobi Optimizer Reference Manual (2025). URL <https://www.gurobi.com>.
- [40] Liang, J. *Benchmarking and Development of Modern Density Functional Theory*. Ph.D. thesis, University of California, Berkeley (2024).
- [41] Liang, J. & Head-Gordon, M. Gold-standard chemical database 137 (gscdb137): A diverse set of accurate energy differences for assessing and developing density functionals. *J. Chem. Theory Comput.* **21**, 12601–12621 (2025).
- [42] Hait, D. & Head-Gordon, M. How accurate is density functional theory at predicting dipole moments? an assessment using a new database of 200 benchmark values. *J. Chem. Theory Comput.* **14**, 1969–1981 (2018). URL <https://doi.org/10.1021/acs.jctc.7b01252>.
- [43] Liu, Y. *et al.* Supervised learning of a chemistry functional with damped dispersion. *Nat. Comput. Sci.* **3**, 48–58 (2023).
- [44] Najibi, A. & Goerigk, L. The nonlocal kernel in van der waals density functionals as an additive correction: An extensive analysis with special emphasis on the b97m-v and ω b97m-v approaches. *J. Chem. Theory Comput.* **14**, 5725–5738 (2018). URL <https://doi.org/10.1021/acs.jctc.8b00842>.
- [45] Lao, K. U. Canonical coupled cluster binding benchmark for nanoscale noncovalent complexes at the hundred-atom scale. *J. Chem. Phys.* **161**, 234103 (2024).
- [46] Galleni, L. *et al.* Benchmarking the GW approximation against coupled-cluster theory for 3d transition metals. *J. Chem. Theory Comput.* **21**, 11620–11631 (2025). URL <https://doi.org/10.1021/acs.jctc.5c01663>.

- [47] Karton, A. A highly diverse and accurate database of 3366 total atomization energies calculated at the ccSD (t)/CBS level by means of w1-f12 theory. *Chem. Phys. Lett.* **868**, 142030 (2025).
- [48] Hoffman, M. P. & Xantheas, S. S. H2X100: A gold-standard CCSD(T)/CBS benchmark dataset of binding energies, structures, and harmonic frequencies for (H2X)_n, X = S, Se, Te, n = 2–4, and assessment of lower-scaling density functional theory methods. *J. Chem. Theory Comput.* **22**, 5539–5554 (2026). URL <https://doi.org/10.1021/acs.jctc.6c00405>.
- [49] Karton, A. & Semidalas, E. Benchmarking isomerization energies for C5–C7 hydrocarbons: The ISOC7 database. *J. Comput. Chem.* **47**, e70296 (2026). URL <https://doi.org/10.1002/jcc.70296>.
- [50] Dahl, R. *et al.* An element-resolved coupled cluster atomization energy data set ranging across the periodic table. *ChemRxiv* (2026). URL <https://doi.org/10.26434/chemrxiv.15005940/v1>. Preprint posted 10 July 2026.
- [51] Hughes, W. B., Popescu, M. V. & Paton, R. S. Fundamental study of density functional theory applied to triplet state reactivity: Introduction of the TRIP50 data set. *J. Chem. Theory Comput.* **22**, 3530–3542 (2026). URL <https://doi.org/10.1021/acs.jctc.6c00144>.
- [52] Karton, A. W1-SN2-BH: A large-scale CCSD(T)/CBS kinetic database. *J. Phys. Chem. A* **130**, 2929–2942 (2026). URL <https://doi.org/10.1021/acs.jpca.6c00424>.
- [53] Knattrup, Y. & Elm, J. Extrapolating local coupled cluster calculations toward CCSD(T)/CBS binding energies of atmospheric molecular clusters. *ACS Omega* **10**, 46794–46808 (2025). URL <https://doi.org/10.1021/acsomega.5c04476>.
- [54] Lao, K. U. Benchmarking nanoscale noncovalent complexes at the two-hundred-atom scale with converged local CCSD(T). *J. Phys. Chem. A* **130**, 4136–4151 (2026). URL <https://doi.org/10.1021/acs.jpca.6c01097>.
- [55] Maier, T. M., Arbuznikov, A. V. & Kaupp, M. Local hybrid functionals: Theory, implementation, and performance of an emerging new tool in quantum chemistry and beyond. *WIREs Comput. Mol. Sci.* **9**, e1378 (2019).
- [56] Luise, G. *et al.* Accurate and scalable exchange-correlation with deep learning. *arXiv preprint arXiv:2506.14665* (2025).